

The Multimodal Anonymizer: a fully local multi-agent AI system for medical data deidentification

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Keywords

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Highlights

- Framework for deidentification of multimodal clinical data.
- Multimodal deidentification with preservation of clinically relevant content.
- On-premises plug-and-play deployment for local data processing.
- Evaluation of 16 model configurations and comparison with existing tools.
- Assessment on external, multilingual and site-specific datasets.

Short Description

The Multimodal Anonymizer is a fully local, multi-agent system that prepares multimodal clinical records for privacy-preserving reuse by coordinating multimodal large language model reasoning, specialist neural networks, rule-based transformations, and iterative verification. Across benchmarks spanning text, tables, PDFs, imaging, metadata, filenames, audio, and handwriting, its best configuration using a local open-source multimodal large language model achieved 98.80% patient-level deidentification sensitivity

and 99.60% preservation of clinically critical content, performing comparably to proprietary models and outperforming established deidentification tools across most modalities.

Abstract

Background: Safe reuse of multimodal hospital data for AI development is limited by the absence of reliable, context-aware deidentification across multimodal data and longitudinal patient data. Existing approaches are largely modality-specific and can indiscriminately remove clinically important information.

Methods: We developed the Multimodal Anonymizer, a modular, locally deployable multi-agent framework integrating multimodal large language models, task-specific neural networks and rule-based transformations. We evaluated 16 orchestrator model configurations on a benchmark built from publicly available data and hospital data from our institution. The benchmark dataset included data from different origins: 250 MIMIC-IV patients with synthetically injected personally identifiable information (PII) supplemented with head CT, face images, handwriting, audio, German clinical-text datasets and local data. Primary outcomes were deidentification sensitivity and preservation of clinically important content; secondary analyses examined model characteristics, reproducibility, and performance against leading market and open-source solutions.

Results: The best local configuration—the orchestrator being Qwen3-VL-235B-A22B-Thinking—achieved near-complete deidentification across all datasets, with per-patient sensitivity of 98.80% (95%-CI 97.20; 100), and per-PII sensitivity of 99.82% (95%-CI 99.76; 99.88). Critical clinical preservation was 99.60% (95%-CI 98.80; 100) per-patient, and clinical preservation was 99.61% (95%-CI 99.51; 99.71) per-file. All modalities achieved at least 98.30% sensitivity (lower bound 95%-CI). On our local data, the system achieved a deidentification sensitivity of 100% per-patient and per-PII; and a critical clinical preservation of 100% per-patient as well as a clinical preservation of 99.97% (95%-CI 99.91; 100) per-file. When comparing orchestrators, the leading local models were similar to proprietary models (GPT-5.2) in deidentification sensitivity while showing higher deidentification specificity. The Multimodal Anonymizer outperformed previous tools on most modalities.

Conclusion: Near-complete, utility-preserving deidentification of multimodal clinical data is achievable with a unified, locally deployable multi-agent system, enabling safer large-scale reuse of hospital data for research and AI development.

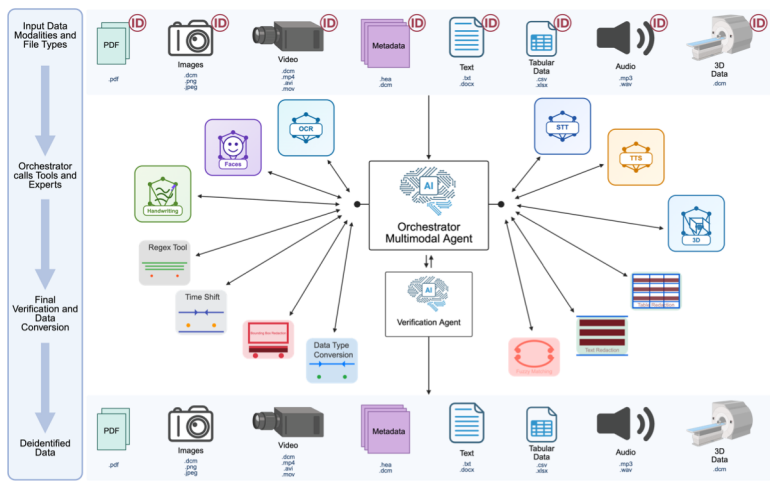
Graphical Abstract

The Multimodal Anonymizer

Multimodal, Locally Deployable Deidentification of Hospital Data

Context-aware multi-agent deidentification across text, images, video, audio, tables, 3D imaging, and real-world clinical data

SYSTEM ARCHITECTURE



KEY RESULTS

Best local configuration

Qwen3-VL-72B-A32B-Thinking as orchestrator

PER-PATIENT SENSITIVITY 98.80% 95% CI 97.20 – 100	CRITICAL CLINICAL PRESERVATION 99.60% 95% CI 98.80 – 100
ALL MODALITIES (LOWER 95% CI) ≥98.30% sensitivity across 10 modalities	CHARITE REAL-WORLD DATA 100% sensitivity and critical preservation

Per-modality sensitivity

Images	99.71%
Faces (2D + 3D)	100%
Tables	99.95%
Audio	100%
PDFs	99.23%
German text	99.94%
Handwriting	99.08%
Text	99.47%
Metadata	100%

KEY TAKEAWAY

Near-complete deidentification with preserved clinical utility, achievable fully on-premises.

Local models match proprietary systems (GPT-5.2) in sensitivity, with higher specificity.

Orchestrator multi-agent framework with rule-based, task-specific and MLLM layers, applied across heterogeneous medical modalities.

Benchmark: 250 MIMIC-IV patients with injected PII + modality-specific datasets + 250 Charité partograms. Comparator: 16 configurations from five model families.

Background

Artificial Intelligence (AI) is transforming many fields and is increasingly reshaping health care. Multimodal models that combine text, images, waveforms, laboratory values, and longitudinal clinical events offer the prospect of substantial quality and efficiency improvements^{1,2}.

To realize that potential, multimodal models must be trained on large, diverse, high-quality datasets. Yet only an estimated 3% of hospital data is currently used, largely because medical data contains direct and indirect patient identifiers³. A critical challenge is the lack of a reliable way to deidentify diverse multimodal data at scale while preserving clinically important information⁴.

Current approaches address only parts of this problem. Named-entity recognition can remove identifiers from text⁵ and deep-learning methods can detect sensitive content in individual modalities⁶, but these methods are typically modality-specific⁴ and cannot interpret information in context⁷—a central requirement when identifiers are embedded across different types of real-world clinical data. Large language models, which can evaluate information in context⁷, have shown promise for deidentifying medical text⁸. However, no context-aware solution exists for multimodal data present ubiquitously in clinical settings.

The next generation of clinical AI will require deidentification systems that are multimodal, scalable, locally deployable, privacy-preserving, and yet simple enough for clinicians without coding expertise to use. Different deidentification aspects also call for different computational strategies: multimodal large language models for contextual reasoning, specialized deep-learning models for modality-specific detection, and rule-based methods for deterministic transformations⁹. Multi-agent systems leverage the advantages of all three aspects while mitigating the flaws^{9,10}.

We developed and evaluated a modular multi-agent system for multimodal deidentification that is easy to use and can run locally within institutional environments. We benchmarked the system against existing approaches and evaluated safety through deidentification sensitivity and clinically critical over-redaction. Our primary hypothesis was that such a system could safely deidentify medical data without over-redacting clinically critical information. Secondary hypotheses were that system performance would improve with orchestrator size, reasoning capability, full-precision deployment, and dense architectures; that the multi-agent design would outperform existing approaches; and that performance would remain stable across repeated runs.

Methods

Study Design and Data Sources

We conducted a retrospective validation study to assess whether a locally deployable multi-agent system could deidentify multimodal clinical data while preserving clinically important information. The automatic evaluation dataset was built from the MIMIC-IV database and linked resources, including core hospital and ICU data¹¹, emergency department records¹², clinical notes¹³, chest radiographs¹⁴, electrocardiograms¹⁵, and echocardiograms¹⁶. Because MIMIC-IV is already deidentified, realistic synthetic personally identifiable information (PII) was injected into all modalities with exact ground-truth annotations. To extend the benchmark beyond data types represented in MIMIC-IV, we added publicly available datasets of volumetric head CT scans^{17,18}, face images¹⁹, handwritten notes²⁰, audio recordings²¹, and synthetic German clinical notes generated by a German physician, with 250 samples per modality assigned one-to-one to the 250 MIMIC-IV-derived patient bundles building our manual evaluation dataset; an overview of the datasets used along with further details is provided in Supplementary S1–S5.

We additionally evaluated the GRASCCO dataset²² as an external German-language text corpus. For site-specific validation, we tested the framework on 250 partograms (gynecologic PDF data) from Charité – Universitätsmedizin Berlin. The study was approved by the Charité Ethics Committee (EA2/267/25).

Benchmark Construction and Ground Truth

Synthetic PII included names, dates of birth, numeric identifiers, and institution-related information. In text and tabular data, MIMIC-IV anonymization placeholders were repopulated with realistic synthetic identifiers and annotated with exact character positions. In image-based data (ECG PDF files, chest radiograph DICOM images, and echocardiogram DICOM images) synthetic identifiers were rendered with varying fonts, sizes, colors, and positions, and their bounding-box coordinates were recorded as ground truth. Free-text injections were generated with model assistance and manually verified before evaluation. Additional modalities not amenable to exact automated labeling were assessed by expert manual review.

Multimodal Anonymizer

We developed the Multimodal Anonymizer as a modular multi-agent framework for multimodal deidentification (Figure 1). A central multimodal large language model (MLLM) orchestrator routes each file to modality-specific processing pipelines. Rule-based methods handled deterministic tasks, including file-type detection and regular-expression-based date detection and shifting. Text-based files were analyzed by the MLLM for PII detection and replacement. For images, PDFs, and videos, the pipeline combined optical character recognition with MLLM analysis to identify and redact

embedded identifiers. Specialized neural networks were used for facial anonymization in volumetric head scans, face detection in photographs, and handwriting recognition. Audio files were transcribed, deidentified as text, and resynthesized. A verification agent iteratively compared the original and processed outputs to detect residual identifiers or unshifted dates and triggered corrective tools when needed. Outputs preserved the original file format and folder structure. Technical implementation details are provided in Supplementary S6–S7.

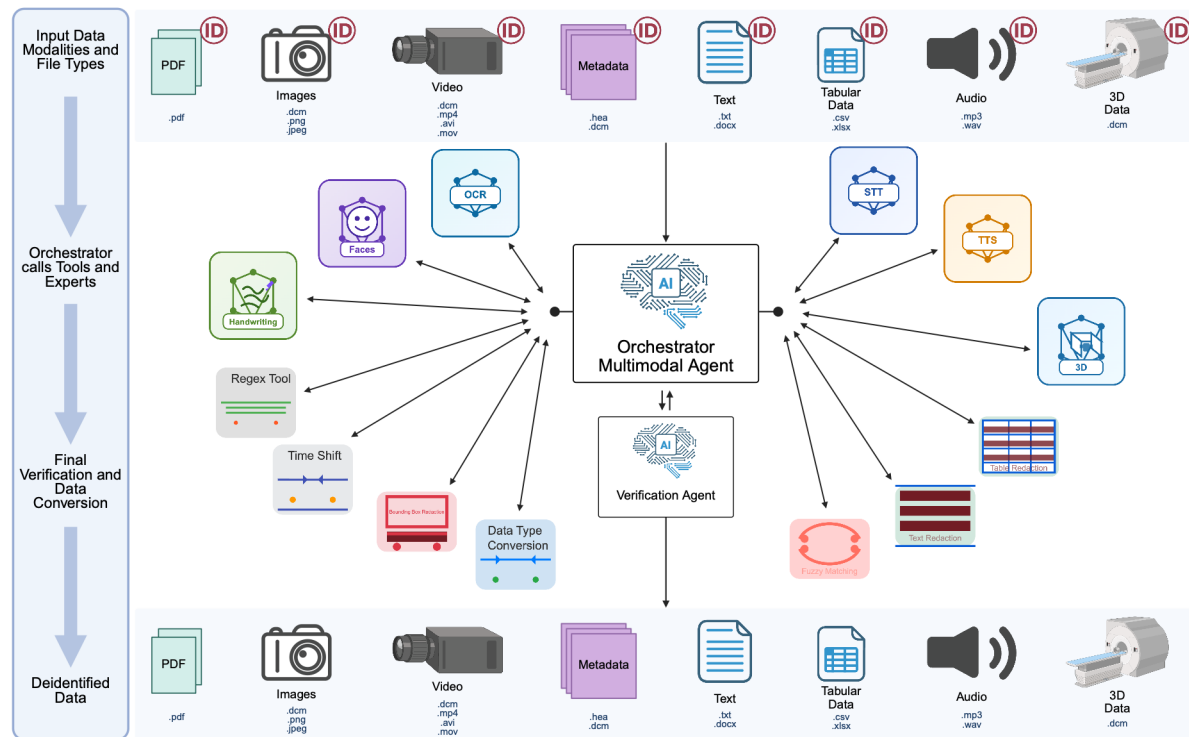


Figure 1: Architecture of the Multimodal Anonymizer

The framework applies modality-specific processing pipelines that are automatically selected based on file type detection. The complete technical specifications are provided in Supplementary S6–S7; here we provide a simplified overview of the key processing steps. The top row shows supported input data modalities and file types, including PDFs, images, videos, metadata, tabular and text-based documents, audio files, and 3D imaging data. As an initial preprocessing step, file and folder names are redacted by the language model, and a patient-specific random time shift (within ± 3 years) is generated and deterministically applied to all dates detected by regular expressions across text-based files. The regex layer covers all common date formats. Non-standard or ambiguous formats missed by regex are handled in the verification step. A central MLLM-based orchestrator then routes each input to specialized tools and models. Text-based data undergo direct language model analysis for PII identification and replacement with standardized placeholders. Image-based data and PDFs combine OCR (EasyOCR²³) with MLLM analysis to detect and redact embedded identifiers, such as patient information on radiographs, using fuzzy matching to account for OCR errors. Specialized neural networks address domain-specific challenges, including facial feature anonymization in neuroimaging data (fine-tuned RetinaNet on self-annotated 3D data¹⁷), face detection in images (fine-tuned ResNet-UNet on an annotated face detection dataset¹⁹), and handwriting recognition (fine-tuned TrOCR^{24,25}

on an annotated handwriting dataset²⁰). For all specialist models, fine-tuning was performed on dedicated training splits that were strictly held out from the evaluation datasets; no evaluation samples were used during model development or prompt engineering. Audio processing integrates speech-to-text transcription (whisper-large-v3²⁶) with text-to-speech synthesis (Kokoro-82M via kokoro-onnx^{27–29}) to produce deidentified recordings. A verification agent iteratively reviews all outputs for missed identifiers or unshifted dates. All pipelines preserve the original file format and return anonymized outputs in the same structure as the inputs (bottom row). CT denotes computed tomography; DICOM, Digital Imaging and Communications in Medicine; MLLM, multimodal large language model; OCR, optical character recognition; PII, personally identifiable information; PDF, portable document format; regex, regular expression; STT, speech-to-text; and TTS, text-to-speech.

Deidentification Standard

The framework implemented deidentification primarily following the 18 HIPAA Safe Harbor identifier categories³⁰. To preserve longitudinal structure, dates were not removed but replaced by a patient-specific consistent random shift applied across that patient's records. Date handling was evaluated separately from identifier removal.

Model Selection and Comparative Evaluation

We evaluated 16 orchestrator model configurations from five families: Kimi-K2.5^{31,32}, GLM-4.6V^{33,34}, Qwen3-VL^{35–42}, Qwen3.5^{43–47}, and GPT-5.2⁴⁸. Open-source models were deployed locally on a cluster with 16 NVIDIA H200 GPUs; GPT-5.2 was accessed through a secured Microsoft Azure deployment as an upper-bound comparator. All models were run with the most deterministic settings available; version details and settings are reported in Supplementary S8. We compared model size, reasoning mode, architecture, and quantization. Across these comparisons, the remaining pipeline components—preprocessing, prompts, and specialist models—were held constant.

For benchmark comparison, we evaluated the Multimodal Anonymizer against existing tools commonly used for deidentification: Presidio⁴⁹ for text-based modalities, ORB-HD^{50,51} for face anonymization, and scitlab's multimodal deidentification tool⁵². Comparator tools were run without task-specific fine-tuning.

Outcomes and Evaluation

The primary outcomes were done by manual review with two independent experts. Disagreements were solved via discussion. Primary outcomes were deidentification sensitivity, critical clinical preservation, and clinical preservation. Deidentification sensitivity was assessed per-patient, defined as the proportion of patients with no residual identifier or combination of identifiers across files that could enable reidentification (see S9), and per-PII entity, defined as the proportion of identifier entities adequately redacted across all datasets. Critical clinical preservation was assessed per-patient and defined as the proportion of patients for whom all clinically essential non-PII information relevant to the current case, such as diagnoses, medications, and treatment plans, remained

unaltered. Clinical preservation was assessed per-file and defined as the proportion of non-PII clinical content within each file that remained unaltered after deidentification.

Secondary outcomes were calculated automatically and included per-PII deidentification sensitivity, per-PII deidentification specificity, per-token false-negative (FNR) and per-token false-positive rate (FPR). Deidentification specificity was defined as the proportion of non-PII content correctly left unredacted. The FNR was the proportion of PII tokens that remained incorrectly unredacted, whereas the FPR was the proportion of non-PII tokens that were incorrectly redacted. Additional outcomes included date-shift performance, reproducibility, and comparative performance against baseline systems.

Automatic evaluation was modality-specific. For text, tables, metadata, and filenames, annotated identifier spans were compared with the redacted output; partially redacted identifiers were counted as failures. For image and PDF modalities, deidentification sensitivity was based on the proportion of annotated bounding-box regions successfully obscured. Date entities were evaluated separately for correct shifting versus redaction or omission.

Statistical Analysis

Patient-level metrics were calculated for each patient bundle. Per-PII, per-file, and per-token metrics were obtained by pooling counts across all datasets. For manually reviewed datasets without annotated true negatives, precision was reported instead of specificity. Uncertainty was estimated with nonparametric bootstrap resampling (10,000 iterations); 95% confidence intervals (CIs) were defined by the 2.5th and 97.5th percentiles. Pairwise model comparisons used paired bootstrap resampling on identical replicate samples and were reported as percentage-point differences (Δ); differences were considered significant when the 95% confidence interval did not include zero. Secondary analyses were exploratory and were not adjusted for multiple comparisons. Reproducibility was assessed across 100 repeated runs of one orchestrator model configuration with identical inputs and settings. All results are reported with bootstrapped 95% CI uncertainty estimates.

Platform Accessibility and Operation

The Multimodal Anonymizer is a fully automated, plug-and-play deidentification pipeline, publicly available on GitHub <https://github.com/charite-iaim/multimodal-anonymizer>. It runs entirely on local infrastructure using user-defined MLLMs, ensuring no patient data leaves institutional servers. Base prompts for all supported modalities are provided but are customizable to accommodate specific requirements.

Documentation guides users through installation and configuration. The pipeline is executed via terminal commands or a browser-based interface in three steps:

1. Configure the local MLLM endpoint and select the model.

2. Upload files with optional prompt customization per modality and filename deidentification settings, including the option to preserve a linkage file connecting deidentified and original data.

3. Download processed, compressed output with original folder structures preserved.

This streamlined interface enables clinical and research personnel to perform comprehensive multimodal deidentification without requiring any programming expertise. Further details on how to use the application, along with example results, can be found in Supplementary S10–S18.

Results

Primary Results

Two independent domain experts manually reviewed all deidentified outputs of the best-performing locally deployed model (Qwen3-VL-235B-A22B-Thinking) across the manual evaluation dataset, assessing both completeness of identifier removal and preservation of clinically relevant content.

The Multimodal Anonymizer achieved near-complete identifier detection, with overall per-patient and per-PII deidentification sensitivities of 98.80% (97.20; 100), and 99.82% (99.76; 99.88), respectively (Figure 2A). For all modalities it achieved at least 98.30% sensitivity (lower bound 95%-CI) (Figure 2B).

Critical clinical preservation was 99.60% (98.80; 100) at the patient level (Figure 2A). A single instance of clinically critical information loss was identified across all datasets (one active pharmaceutical ingredient was redacted from a discharge note). General clinical preservation, which measures whether clinical information unrelated to the patient's case was inadvertently removed, was 99.61% (99.51; 99.71) and exceeded 97.19% (lower bound 95%-CI) across all modalities (Figure 2C). Full results are available in Supplementary S19.

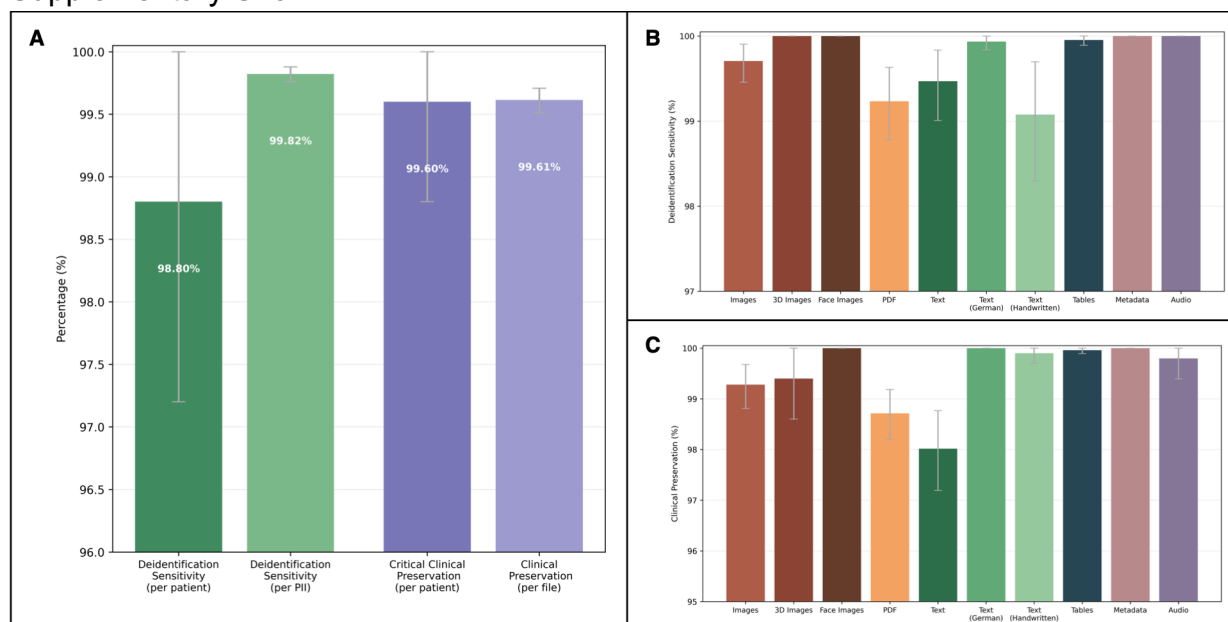


Figure 2: Manual evaluation of deidentification sensitivity, critical clinical preservation, and clinical preservation of the best-performing locally deployed model (Qwen3-VL-235B-A22B-Thinking).

Deidentification sensitivity and (critical) clinical preservation are reported at multiple aggregation levels. Deidentification sensitivity is reported per-patient (proportion of patients without any critical omission) and per-PII entity (proportion of all identifier entities across all datasets that were adequately redacted). Per-patient deidentification sensitivity reflects whether any combination of residual identifiers across a patient's files could enable reidentification; a patient is counted as adequately

deidentified only if no such combination remains (see S9). Critical clinical preservation denotes the proportion of patients for whom all clinically highly relevant non-PII content pertaining to the current case, such as diagnoses, medications, and treatment plans, remained unaltered after deidentification. Clinical preservation is reported per-file and denotes the proportion of all non-PII clinical content (including less critical items such as pharmaceutical manufacturer names) that remained unaltered, pooled across all patients. All panels show 95% confidence intervals. **Panel A** shows overall deidentification sensitivity and (critical) clinical preservation. **Panel B** shows per-PII deidentification sensitivity stratified by modality, and **Panel C** shows per-file clinical preservation stratified by modality. PII denotes personally identifiable information.

To evaluate generalizability on a local German Charité dataset, we performed an identical analysis on partograms from 250 patients. The system achieved a deidentification sensitivity of 100% at both per-patient and per-PII aggregation levels, indicating complete removal of all identifiable information across the Charité dataset (Table 1). Critical clinical preservation was likewise 100% per-patient, with no clinically critical information redacted. General clinical preservation was 99.97% (99.91; 100) per-file.

Table 1. Deidentification Sensitivity, Critical Clinical Preservation and Clinical Preservation (local Charité data).*

Metric (± 95% CI)	Model Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking)
Deidentification Sensitivity (per-patient)	100
Deidentification Sensitivity (per-PII)	100
Critical Clinical Preservation (per-patient)	100
Clinical Preservation (per-file)	99.97 ± (99.91, 100)

* All values are percentages with 95% bootstrapped CIs. CI denotes confidence interval; and PII, personally identifiable information.

Secondary Results

Complete metrics and results of pairwise model comparison are provided in Supplementary S20–S22.

Performance across models

Table 2 summarizes the deidentification evaluation on the automatic evaluation dataset for the five largest models (see Supplementary S23–S26 for visualizations).

Table 2. Evaluation of Deidentification Performance on the automatic evaluation dataset. *

Modality and Metric (± 95% CI)	GPT-5.2	Kimi-K2.5 (thinking)	Model Qwen3-VL-235B-A22B-Thinking	Qwen3.5-397B-A17B (thinking)	GLM-4.6V
Overall					
Sensitivity	99.98 ± (99.96, 99.99)	98.97 ± (98.56, 99.33)	99.91 ± (99.83, 99.96)	99.92 ± (99.88, 99.95)	98.90 ± (97.92, 99.56)
Specificity	77.37 ± (76.23, 78.52)	95.22 ± (93.77, 96.26)	94.95 ± (93.46, 96.01)	95.08 ± (93.61, 96.12)	92.84 ± (91.01, 94.30)
FNR (per-token)	0.01 ± (0.00, 0.02)	0.41 ± (0.30, 0.55)	0.08 ± (0.05, 0.12)	0.13 ± (0.08, 0.21)	2.18 ± (0.88, 4.00)
FPR (per-token)	11.48 ± (10.52, 12.71)	2.51 ± (1.66, 3.79)	2.62 ± (1.76, 3.90)	2.61 ± (1.75, 3.91)	4.31 ± (3.01, 5.94)
Table					

Sensitivity	99.98 ± (99.95, 99.99)	99.96 ± (99.91, 100)	99.95 ± (99.90, 99.99)	99.90 ± (99.73, 99.99)	95.68 ± (91.22, 98.89)
Specificity	76.21 ± (74.73, 77.47)	80.71 ± (77.54, 83.61)	81.04 ± (77.82, 83.90)	80.43 ± (77.12, 83.46)	78.99 ± (75.57, 82.20)
FNR (per-token)	0.01 ± (0.00, 0.02)	0.03 ± (0.00, 0.07)	0.03 ± (0.00, 0.07)	0.08 ± (0.00, 0.22)	3.84 ± (1.00, 7.71)
FPR (per-token)	12.69 ± (11.45, 14.49)	7.86 ± (5.28, 11.83)	7.68 ± (5.11, 11.61)	8.14 ± (5.49, 12.09)	9.43 ± (6.61, 13.48)
Text					
Sensitivity	98.78 ± (97.80, 99.60)	99.34 ± (98.94, 99.69)	99.41 ± (99.01, 99.75)	98.90 ± (98.39, 99.37)	95.25 ± (94.14, 96.31)
Specificity	92.88 ± (86.90, 97.25)	97.41 ± (95.72, 98.52)	97.03 ± (95.31, 98.20)	97.28 ± (95.59, 98.39)	94.65 ± (92.43, 96.42)
FNR (per-token)	1.20 ± (0.38, 2.19)	0.56 ± (0.25, 0.92)	0.50 ± (0.21, 0.85)	0.99 ± (0.58, 1.44)	4.70 ± (3.58, 5.88)
FPR (per-token)	6.38 ± (1.97, 12.34)	1.99 ± (0.89, 3.67)	2.15 ± (1.04, 3.83)	2.09 ± (0.98, 3.77)	4.33 ± (2.57, 6.53)
Image					
Sensitivity	99.89 ± (99.87, 99.91)	98.45 ± (98.11, 98.77)	99.65 ± (99.50, 99.77)	99.72 ± (99.60, 99.82)	99.58 ± (99.42, 99.72)
Specificity	99.29 ± (99.27, 99.31)	99.19 ± (99.13, 99.26)	99.17 ± (99.10, 99.24)	99.22 ± (99.15, 99.29)	99.20 ± (99.13, 99.27)
FNR (per-token)	0.13 ± (0.10, 0.15)	1.73 ± (1.28, 2.29)	0.31 ± (0.21, 0.42)	0.38 ± (0.24, 0.57)	0.56 ± (0.35, 0.83)
FPR (per-token)	0.71 ± (0.69, 0.73)	0.81 ± (0.74, 0.87)	0.83 ± (0.76, 0.90)	0.78 ± (0.71, 0.85)	0.80 ± (0.73, 0.87)
PDF					
Sensitivity	99.97 ± (99.89, 100)	99.20 ± (98.92, 99.46)	99.89 ± (99.80, 99.97)	99.96 ± (99.91, 100)	99.82 ± (99.69, 99.92)
Specificity	95.51 ± (95.41, 95.62)	95.72 ± (95.66, 95.78)	95.72 ± (95.67, 95.78)	95.78 ± (95.73, 95.83)	95.76 ± (95.71, 95.82)
FNR (per-token)	0.05 ± (0.00, 0.17)	1.21 ± (0.80, 1.65)	0.21 ± (0.04, 0.45)	0.05 ± (0.00, 0.11)	0.30 ± (0.13, 0.51)
FPR (per-token)	4.49 ± (4.38, 4.59)	4.28 ± (4.22, 4.34)	4.28 ± (4.22, 4.33)	4.22 ± (4.17, 4.27)	4.24 ± (4.18, 4.29)
Metadata					
Sensitivity	100	100	100	100	99.88 ± (99.78, 99.97)
Specificity	97.16 ± (96.42, 97.89)	98.20 ± (98.02, 98.34)	98.39 ± (98.39, 98.39)	97.97 ± (97.69, 98.21)	98.25 ± (98.07, 98.39)
FNR (per-token)	0.00	0.00	0.00	0.00	0.12 ± (0.03, 0.22)
FPR (per-token)	2.03 ± (1.59, 2.52)	1.33 ± (1.24, 1.43)	1.24 ± (1.22, 1.26)	1.44 ± (1.32, 1.59)	1.34 ± (1.25, 1.45)
Filenames					
Sensitivity	100	98.47 ± (97.80, 99.07)	99.94 ± (99.81, 100)	100	100
Specificity	98.31 ± (98.22, 98.40)	99.71 ± (99.56, 99.85)	98.81 ± (98.45, 99.15)	100	99.86 ± (99.79, 99.92)
FNR (per-token)	0.00	1.53 ± (0.93, 2.20)	0.06 ± (0.00, 0.19)	0.00	0.00
FPR (per-token)	1.69 ± (1.60, 1.78)	0.29 ± (0.15, 0.44)	1.19 ± (0.86, 1.56)	0.00	0.14 ± (0.08, 0.21)
Date Shift					
Shift Rate	98.66 ± (98.42, 98.86)	99.64 ± (99.42, 99.81)	99.40 ± (98.81, 99.78)	99.63 ± (99.41, 99.80)	98.04 ± (96.94, 98.94)
Redaction Rate	1.21 ± (1.02, 1.45)	0.09 ± (0.02, 0.18)	0.27 ± (0.00, 0.80)	0.05 ± (0.01, 0.12)	1.30 ± (0.56, 2.25)
FNR (per-token)	0.02 ± (0.01, 0.03)	0.00 ± (0.00, 0.01)	0.00 ± (0.00, 0.01)	0.00 ± (0.00, 0.01)	0.01 ± (0.00, 0.04)

* Sensitivity and specificity refer to deidentification sensitivity and deidentification specificity, respectively. Per-token FNR and FPR quantify residual exposure at the character level (text modalities) or spatial bounding-box level (images, PDF). Each column reflects a different model used as the MLLM for the orchestrator. All values are percentages with 95% bootstrapped CIs. CI denotes confidence interval; FNR, false-negative rate; FPR, false-positive rate; and PII, personally identifiable information.

Overall deidentification sensitivity was highest for GPT-5.2, followed closely by Qwen3.5-397B-A17B-thinking and Qwen3-VL-235B-A22B-Thinking. GPT-5.2 performed marginally but significantly better than Qwen3.5-397B-A17B-thinking (Δ 0.06% $\pm$ 0.03; 0.10) and Qwen3-VL-235B-A22B-Thinking (Δ 0.07% $\pm$ 0.01; 0.16), although the magnitude of these differences was small. GPT-5.2 achieved significantly better results than Kimi-K2.5-thinking (Δ 1.01% $\pm$ 0.64; 1.41) or GLM-4.6V (Δ 1.08% $\pm$ 0.43; 2.07).

Deidentification specificity was highest for Kimi-K2.5-thinking, followed by Qwen3.5-397B-A17B-thinking and Qwen3-VL-235B-A22B-Thinking. Kimi-K2.5-thinking did not differ significantly from either Qwen3.5-397B-A17B-thinking (Δ 0.15% $\pm$ -1.68; 1.93) or Qwen3-VL-235B-A22B-Thinking (Δ 0.28% $\pm$ -1.38; 1.97).

GPT-5.2 showed the lowest deidentification specificity, the second-worst model was GLM-4.6V. GPT-5.2 was significantly worse than GLM-4.6V for deidentification specificity (Δ -15.50% $\pm$ -17.41; -13.38).

The best-performing local model (Qwen3-VL-235B-A22B-Thinking) achieved deidentification sensitivity with lower confidence bounds exceeding 99% in every

modality. Notably, it outperformed GPT-5.2 in date shifting accuracy ($\Delta 4.45\% \pm 3.43; 5.52$), with fewer inadvertent date redactions ($\Delta -0.95\% \pm -1.37; -0.34$).

Model scale

Across the Qwen3-VL-Instruct (4B–235B) and Qwen3.5 (9B–397B) families, deidentification sensitivity performance improved with model size but with diminishing returns (see S27–S30). Pairwise bootstrap analysis confirmed a significant gain from the 8B to the 32B Qwen3-VL-Instruct variant ($\Delta 0.19\% \pm 0.09; 0.29$). Qwen3.5, by contrast, already showed no significant difference between 9B and 27B ($\Delta -0.03\% \pm -0.09; 0.03$). No major differences emerged at the top of each family for Qwen3-VL-Instruct 30B-A3B vs 235B-A22B ($\Delta 0.15\% \pm -0.03; 0.32$) and Qwen3.5 35B-A3B vs 397B-A17B ($\Delta 0.03\% \pm -0.05; 0.16$), except for date shifting accuracy, which improved progressively with scale: smaller models tended to redact dates instead of preserving them as instructed.

Reasoning capability

Enabling reasoning mode significantly improved overall deidentification sensitivity for Qwen3-VL-235B-A22B ($\Delta 0.37\% \pm 0.20; 0.56$), had no effect on Qwen3.5-397B-A17B ($\Delta 0.05\% \pm -0.03; 0.18$), and worsened the deidentification sensitivity of Kimi-K2.5 ($\Delta -0.91\% \pm -1.31; -0.53$) (see S31–S33).

Architecture

The dense Qwen3-VL-32B-Instruct and the similarly sized mixture-of-experts Qwen3-VL-30B-A3B-Instruct did not differ significantly in deidentification sensitivity ($\Delta 0.11\% \pm -0.05; 0.35$) or specificity ($\Delta -0.92\% \pm -2.72; 0.95$) (see S34–S36).

Quantization

Four-bit quantization (AWQ) of Qwen3-VL-32B-Instruct had no significant effect on deidentification sensitivity ($\Delta -0.05\% \pm -0.11; 0.01$) or specificity ($\Delta -0.26\% \pm -0.88; 0.34$) (see S37–S39).

Reproducibility

One hundred independent runs of Kimi-K2.5 demonstrated high stability, with narrow confidence intervals across all modalities and metrics (see S40). Text deidentification sensitivity showed the largest variation ($98.77\% \pm 98.11; 99.11$); all other metrics showed smaller 95% CIs.

Benchmark Comparison

We compared the Multimodal Anonymizer using Qwen3-VL-235B-A22B-Thinking against commonly used deidentification tools: Microsoft’s Presidio⁴⁹, the face-redaction library ORB-HD⁵⁰ based on CenterFace⁵¹, and a multimodal deidentification tool developed by scitlab⁵² (Table 3). Pairwise bootstrap comparison results are provided in Supplementary S41 and S42.

The Multimodal Anonymizer significantly outperformed Presidio ($\Delta 4.61\% \pm 3.91$; 5.30) and scitlab ($\Delta 27.18\% \pm 26.60$; 27.75) for deidentification sensitivity on tabular data. Furthermore, it outperformed Presidio ($\Delta 61.24\% \pm 57.86$; 64.58) and scitlab ($\Delta 23.62\% \pm 23.08$; 24.04) for deidentification sensitivity on imaging data.

On free text, deidentification sensitivity was comparable to Presidio ($\Delta 0.01\% \pm -0.61$; 0.65), while specificity was substantially higher ($\Delta 72.90\% \pm 68.14$; 77.46), indicating extensive over-redaction by Presidio. In contrast, scitlab showed significantly lower text sensitivity ($\Delta 10.39\% \pm 9.62$; 11.14) with comparable specificity ($\Delta 2.24\% \pm -0.52$; 4.86). Presidio did not support metadata and filename deidentification. scitlab showed substantially lower deidentification sensitivity for metadata ($\Delta -81.64\% \pm -82.96$; -80.23) and filenames ($\Delta -15.12\% \pm -15.48$; -14.75). Date shifting was not supported by scitlab; instead, it redacted 94.75% (90.89; 97.95) of dates.

The Multimodal Anonymizer significantly outperformed Presidio in deidentification sensitivity on handwritten notes ($\Delta 47.90\% \pm 43.30$; 52.40), German letters ($\Delta 23.40\% \pm 22.50$; 24.30), and the GRASCCO dataset ($\Delta 12.50\% \pm 8.30$; 17.00). Face detection sensitivity was comparable to ORB-HD ($\Delta 0.80\% \pm 0.00$; 2.10).

Table 3. Performance metrics for the benchmark comparison against existing deidentification tools.*

Modality and Metric ($\pm 95\%$ CI)	Model			
	Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking)	Presidio	ORB-HD	scitlab
Table				
Sensitivity	99.95 $\pm$ (99.90, 99.99)	93.17 $\pm$ (91.30, 94.86)	-	47.74 $\pm$ (45.79, 50.54)
Specificity	81.04 $\pm$ (77.82, 83.90)	46.59 $\pm$ (37.85, 56.49)	-	78.33 $\pm$ (77.32, 79.39)
FNR (per-token)	0.03 $\pm$ (0.00, 0.07)	4.86 $\pm$ (3.86, 5.91)	-	43.16 $\pm$ (41.27, 44.52)
FPR (per-token)	7.68 $\pm$ (5.11, 11.61)	70.21 $\pm$ (62.39, 76.61)	-	18.82 $\pm$ (17.70, 20.24)
Text				
Sensitivity	99.41 $\pm$ (99.01, 99.75)	99.49 $\pm$ (99.08, 99.82)	-	90.75 $\pm$ (89.94, 91.53)
Specificity	97.03 $\pm$ (95.31, 98.20)	20.08 $\pm$ (16.55, 23.93)	-	94.73 $\pm$ (92.15, 96.85)
FNR (per-token)	0.50 $\pm$ (0.21, 0.85)	0.57 $\pm$ (0.20, 1.03)	-	10.49 $\pm$ (9.68, 11.31)
FPR (per-token)	2.15 $\pm$ (1.04, 3.83)	83.00 $\pm$ (79.52, 86.21)	-	7.30 $\pm$ (5.18, 9.89)
Image				
Sensitivity	99.65 $\pm$ (99.50, 99.77)	39.40 $\pm$ (36.15, 42.62)	-	75.89 $\pm$ (75.60, 76.17)
Specificity	99.17 $\pm$ (99.10, 99.24)	99.77 $\pm$ (99.72, 99.81)	-	99.88 $\pm$ (99.85, 99.91)
FNR (per-token)	0.31 $\pm$ (0.21, 0.42)	59.34 $\pm$ (56.05, 62.60)	-	22.33 $\pm$ (22.03, 22.64)
FPR (per-token)	0.83 $\pm$ (0.76, 0.90)	0.23 $\pm$ (0.19, 0.28)	-	0.12 $\pm$ (0.09, 0.15)
PDF				
Sensitivity	99.89 $\pm$ (99.80, 99.97)	93.75 $\pm$ (92.78, 94.69)	-	97.72 $\pm$ (97.41, 97.99)
Specificity	95.72 $\pm$ (95.67, 95.78)	99.63 $\pm$ (99.61, 99.64)	-	94.97 $\pm$ (94.93, 95.02)
FNR (per-token)	0.21 $\pm$ (0.04, 0.45)	8.17 $\pm$ (6.90, 9.48)	-	2.54 $\pm$ (2.23, 2.87)
FPR (per-token)	4.28 $\pm$ (4.22, 4.33)	0.37 $\pm$ (0.36, 0.39)	-	5.03 $\pm$ (4.98, 5.07)
Metadata				
Sensitivity	100	-	-	18.37 $\pm$ (17.03, 19.74)
Specificity	98.39 $\pm$ (98.39, 98.39)	-	-	97.74 $\pm$ (97.66, 97.82)
FNR (per-token)	0.00	-	-	81.63 $\pm$ (80.26, 82.97)
FPR (per-token)	1.24 $\pm$ (1.22, 1.26)	-	-	2.98 $\pm$ (2.92, 3.05)
Filenames				
Sensitivity	99.94 $\pm$ (99.81, 100)	-	-	77.13 $\pm$ (76.63, 77.62)
Specificity	98.81 $\pm$ (98.45, 99.15)	-	-	87.50 $\pm$ (87.29, 87.71)
FNR (per-token)	0.06 $\pm$ (0.00, 0.19)	-	-	22.87 $\pm$ (22.38, 23.37)

FPR (per-token)	1.19 ± (0.86, 1.56)	-	-	12.50 ± (12.29, 12.71)
Date Shift				
Shift Rate	99.40 ± (98.81, 99.78)	-	-	-
Redaction Rate	0.27 ± (0.00, 0.80)	-	-	94.75 ± (90.89, 97.95)
FNR (per-token)	0.00 ± (0.00, 0.01)	-	-	0.00
Face Images				
Sensitivity	100	-	99.22 ± (97.90, 100)	-
Precision	91.99 ± (88.60, 95.04)	-	95.04 ± (92.62, 97.26)	-
Text (Handwriting)				
Sensitivity	99.08 ± (98.31, 99.70)	51.17 ± (46.81, 55.52)	-	-
Precision	90.32 ± (87.99, 92.57)	93.85 ± (90.88, 96.53)	-	-
Text (German)				
Sensitivity	99.94 ± (99.84, 100)	76.55 ± (75.66, 77.43)	-	n.e.
Precision	100	53.15 ± (52.42, 53.89)	-	n.e.
GRASCCO				
Sensitivity	96.77 ± (94.70, 98.61)	84.24 ± (80.33, 87.88)	-	n.e.
Precision	99.17 ± (97.32, 100)	17.31 ± (14.35, 20.97)	-	n.e.

* The Multimodal Anonymizer using Qwen3-VL-235B-A22B-Thinking was compared against Presidio (v2.2.361), ORB-HD (v1.5.0), and scitlab (version as of February 2025) across all supported modalities. None of the comparator tools were fine-tuned for the benchmark data. Results for tabular data, free text, images, PDFs, metadata, filenames, and date shifting were obtained through automated evaluation against ground truth annotations. Face images, handwritten text, German text, and GRASCCO results were assessed by two independent domain experts. True negatives were unavailable for manually evaluated data, therefore precision is reported instead of specificity. Neither Presidio nor scitlab supported the full modality range: dashes indicate unsupported modalities; "n.e." indicates a modality that the tool supports but was not evaluated in this benchmark. Sensitivity and specificity refer to deidentification sensitivity and deidentification specificity, respectively. All values are percentages with 95% bootstrapped CIs. CI denotes confidence interval; FNR, false-negative rate; FPR, false-positive rate; and n.e., not evaluated.

Discussion

In this study, we developed and evaluated a modular, locally deployable framework for deidentification and found that near-complete identifier removal across multimodal clinical data types can be achieved with minimal loss of clinically important information. The framework showed strong performance across text, tables, PDFs, images, metadata, filenames, audio, 3D images and, in most comparisons, outperformed and supported more modalities than available alternatives. These findings suggest that multimodal deidentification is feasible within governed institutional environments and may lower a major barrier to the secondary use of hospital data for research and AI development^{8,53}. The importance of these findings is practical as much as technical. In multimodal clinical AI, the central bottleneck is often not model architecture but the inability to safely transform routine hospital data into reusable research assets⁵⁴. Identifiers are distributed across document bodies, headers, overlays, metadata, filenames, and temporal fields, and no single method is likely to be sufficient. Our results support a systems view of deidentification: deterministic rules, specialist models, MLLM reasoning, and verification steps each contribute distinct strengths, and their integration appears more effective than relying on any one approach alone⁹.

Consistent with the need to balance disclosure risk and data utility^{55,56}, we evaluated deidentification not by sensitivity alone, but jointly with preservation of clinically meaningful information. In clinical records, utility depends not only on retaining individual facts such as diagnoses, medications, and treatment plans, but also on preserving longitudinal temporal structure. In our comparisons, the best local configuration achieved high identifier removal while limiting unnecessary redaction of clinically important content and preserving temporal structure through date shifting more reliably than comparator systems. Thus, our contribution is the operationalization and empirical assessment of this trade-off for multimodal clinical data deidentification.

Patient-level evaluation was particularly important. In practice, disclosure risk is cumulative: a residual name in one file, an unshifted date in another, and a revealing filename or metadata field elsewhere may together enable reidentification even when document-level performance appears strong. Assessing complete multimodal patient records therefore better reflects operational risk than token- or document-level metrics alone. The same principle applies to utility, because small errors distributed across multiple files can distort the longitudinal clinical narrative⁵⁷.

These findings also have implications for deployment. Strong performance from a local model is important because health systems often require data residency, auditability, and institutional control⁷. A unified workflow is also more likely to be adopted than a fragmented collection of modality-specific tools^{8,58}. At the same time, date shifting illustrates that deidentification remains a governed trade-off: preserving within-patient temporal structure can substantially improve scientific usefulness, but it may not be

appropriate for every data deidentification context and should be paired with setting-specific oversight³⁰.

Our study has several limitations: (1) The benchmark relied substantially on MIMIC-IV with synthetic identifier injection, which enabled precise ground truth but may not fully capture the variability of naturally occurring identifiers in routine clinical care. Validation on Charité data confirmed strong performance on naturally occurring identifiers, but generalizability across institutions, languages, specialties, and document types warrants further validation. (2) Residual reidentification risk cannot be fully eliminated, particularly through linkage of rare clinical features or external data sources^{30,59}. (3) Best performance required substantial computational resources. (4) Data utility was inferred from preservation metrics rather than tested directly in downstream modeling tasks.

In summary, high-performing, locally deployable multimodal deidentification appears achievable across diverse clinical data types with minimal loss of clinically important information. More broadly, the study suggests that deidentification can be operationalized as a unified institutional workflow rather than a collection of isolated tools. If validated prospectively across institutions, systems of this kind could become foundational infrastructure for the safe reuse of multimodal clinical data in research and AI development.

Reporting, Open Science and Ethical Approval

This retrospective benchmark-development and validation study is reported with detailed descriptions of study design, benchmark construction, model evaluation, and statistical analysis in the Methods and Supplementary. Publicly available datasets were used in accordance with their respective access and licensing terms. The Charité Ethics Committee approved the retrospective use and anonymization of the institution-specific partogram dataset from Charité – Universitätsmedizin Berlin (EA2/267/25) without requiring individual informed consent. Reproducibility materials are provided as permitted by data-use agreements, institutional policy, and patient privacy requirements.

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Authorship Contributions

AH contributed to the study design, defined research questions, performed AI Inference and statistical analysis, wrote the draft of the manuscript and should be considered first author. FWT and KSK performed AI inference. RG, TR, MG, RE and MS provided methodological guidance and contributed to the formal analysis. PR provided the Charité data used for analysis. FP designed the study, provided methodological supervision and the technical context of the study. AM designed the study, provided methodological supervision and the technical context of the study and should be considered last author. JM designed the study, defined research questions, performed AI Inference and statistical analysis, wrote the draft of the manuscript and should be considered last author. All authors contributed to revising the manuscript prior to submission.

Data Availability Statement

The data analyzed in this study were obtained from a combination of credentialed-access, publicly available, and institution-specific sources. MIMIC-IV and linked MIMIC-IV resources are available from their original providers subject to required training, data-use agreements, and credentialed access. Additional public datasets are available from their original sources under their respective licenses and access conditions. The institution-specific partogram dataset from Charité – Universitätsmedizin Berlin is not publicly available because it contains sensitive clinical information and is subject to legal and

institutional restrictions. Derived benchmark annotations, synthetic-PII injection specifications and aggregate evaluation outputs are available from the corresponding author on reasonable request, where consistent with applicable data-use agreements and privacy regulations.

Code Availability Statement

The code used for benchmark construction, evaluation, and statistical analysis is available at <https://github.com/charite-iaim/multimodal-anonymizer>. A plug-and-play version for non-technical user is provided to reproduce the reported analyses, together with documentation, prompts, model settings, and package versions in the code repository. The framework is provided with an easy-to-use browser-based user interface in order to facilitate the wide adoption of the framework.

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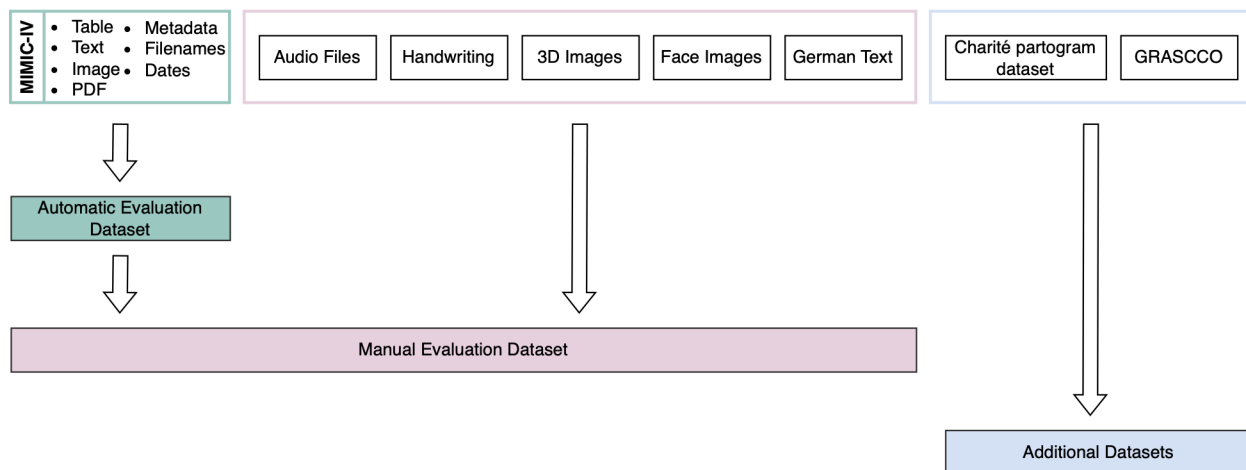
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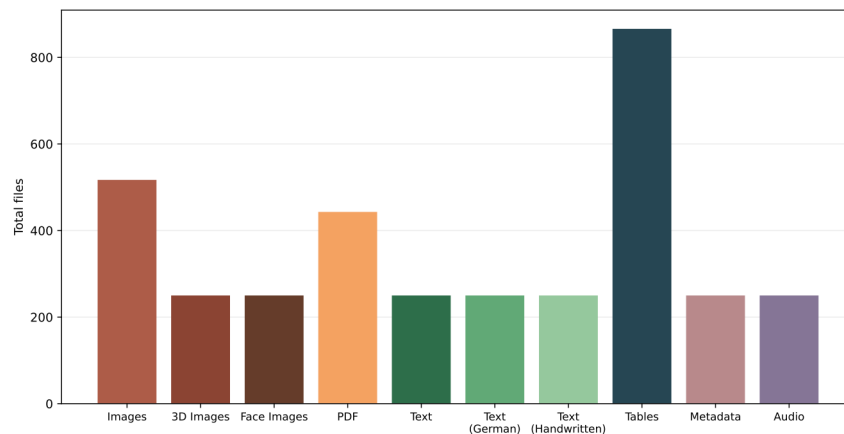
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Supplementary

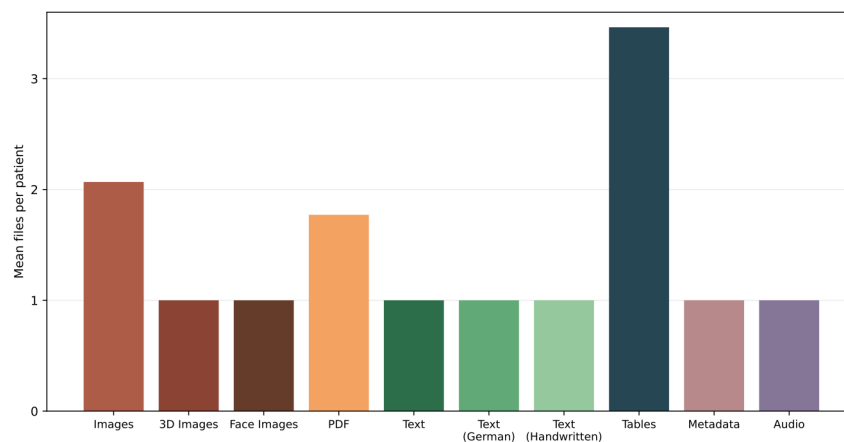


Supplementary Figure 1: Construction of the Evaluation Datasets. All datasets contained PII except for MIMIC-IV. We therefore systematically injected synthetic identifiers into the MIMIC-IV dataset across all modalities to create an annotated dataset for deidentification evaluation. Electrocardiogram data underwent dual processing. To evaluate performance on the common clinical scenario of scanned or printed ECG reports, we converted the time-series waveform data from its native format into PDF documents. Each PDF was injected with a synthetic patient name, date of birth, and patient identifier, along with additional randomly selected fields. These included further PII such as physician name, study date, and institution name, as well as non-PII values such as heart rate, PR interval, and QRS duration. Identifiers were placed at several randomly selected positions within each file to ensure that the framework

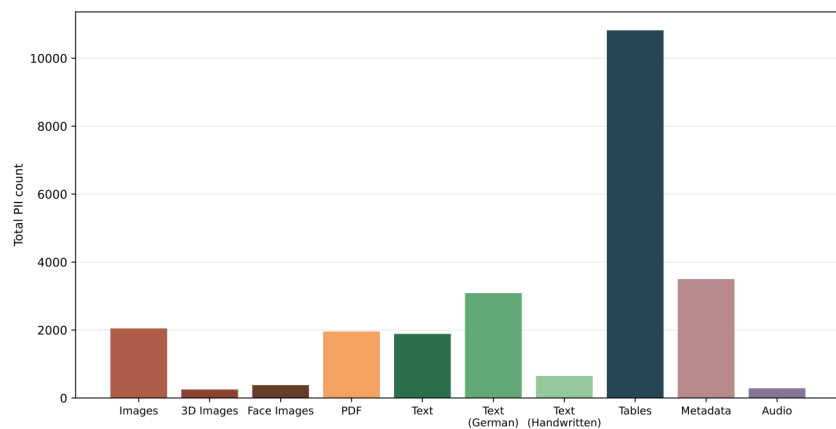
detects information in different locations. ECG header metadata files were annotated separately; their clearly structured format allowed precise specification of identifier locations for subject identifiers and date information. Chest radiographs as well as echocardiogram videos and images in DICOM format received identifier injections analogous to those used for the ECGs. Bounding box coordinates were recorded for all injected identifiers, enabling the PII to be localized for subsequent evaluation. Tabular data presented unique challenges due to the complexity and volume of clinical database exports. We derived CSV files from the MIMIC-IV database structure. Annotation followed an XML-style scheme, with personal identifiers marked by <PER> tags and temporal information by <DATE> tags. Columns containing PIIs, such as subject and admission identifiers, were annotated by explicitly defining which columns constitute PIIs. Where appropriate, we added further PII columns containing, among other things, names, social security numbers, and addresses. Free-text columns, including clinical comments, underwent combined annotation and injection using GPT-5.1. Discharge summaries and radiology notes received an enriched injection of identifiers such as names, locations, and telephone numbers, with all generated annotations subsequently verified and supplemented manually. The textual data used for evaluation consisted of the annotated and injected discharge-summary text columns, exported as standalone text files. Finally, file and folder names throughout the dataset were annotated for any embedded identifiers using <PER> tags, allowing us to assess whether personal identifiers in filenames are correctly redacted.



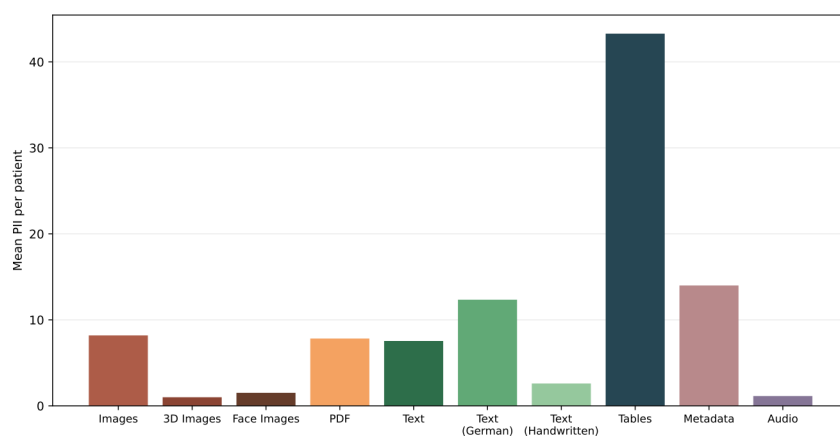
Supplementary Figure 2: Distribution of files per modality in the evaluation dataset.



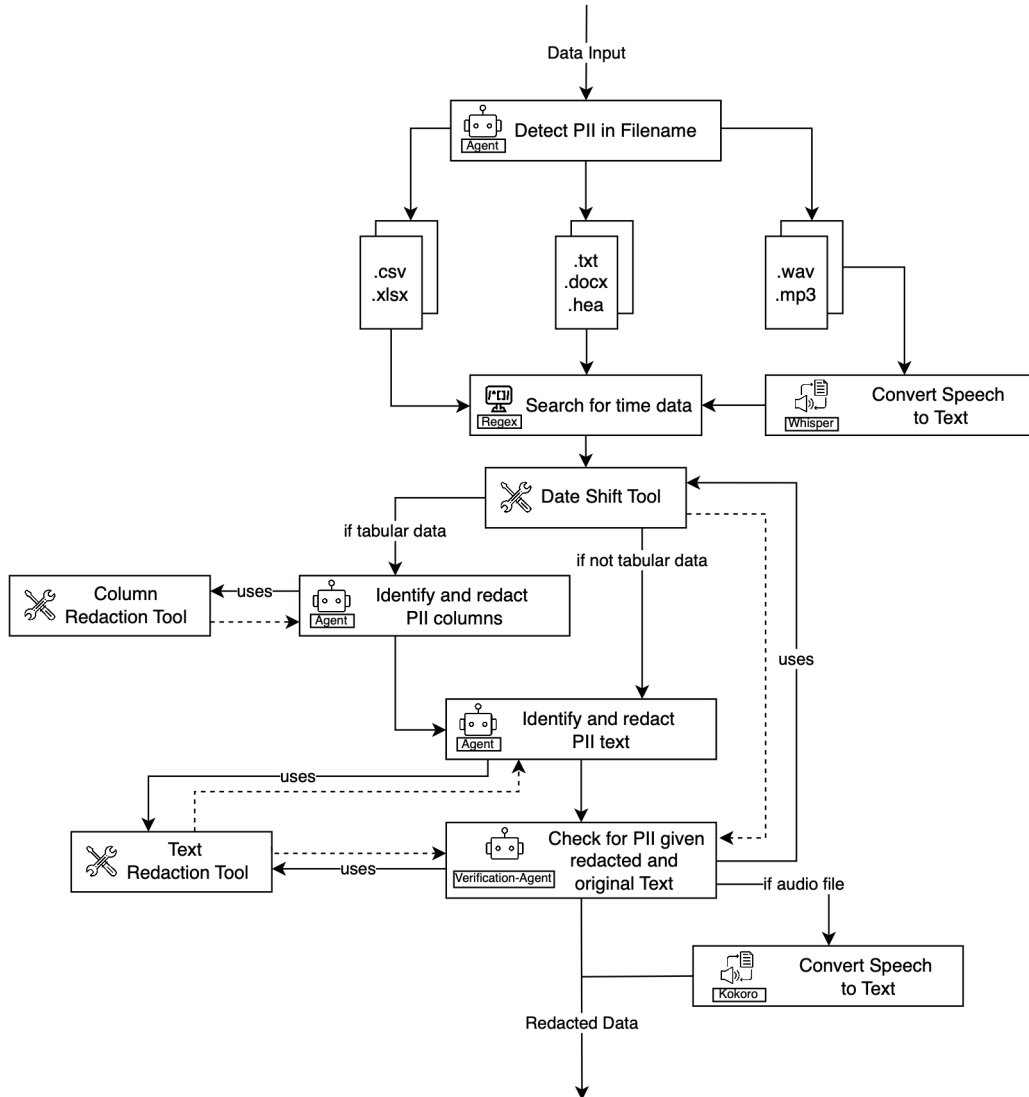
Supplementary Figure 3: Mean number of files per patient across modalities in the evaluation dataset.



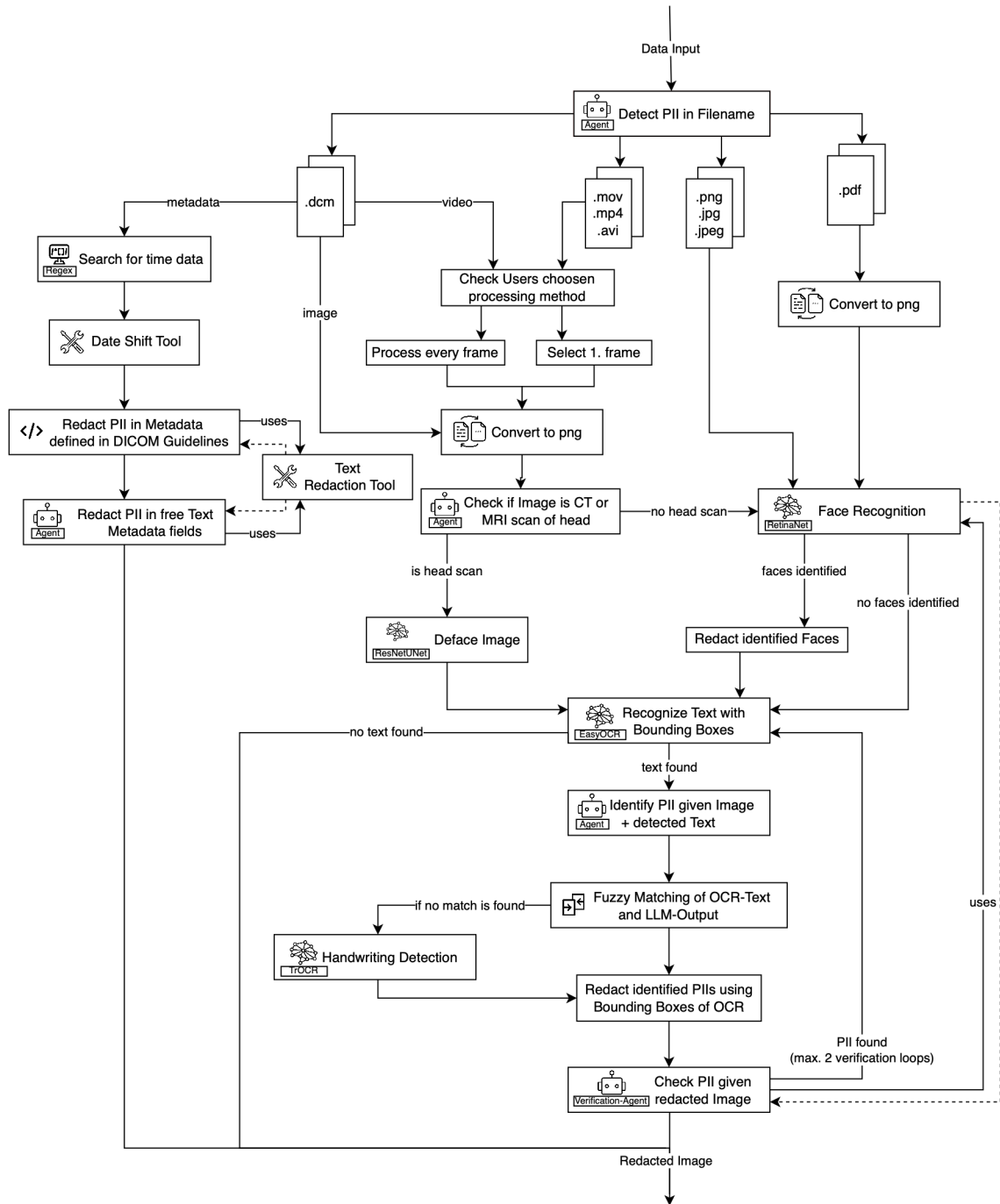
Supplementary Figure 4: Total count of personal identifiable information (PII) instances across modalities in the evaluation dataset.



Supplementary Figure 5: Mean number of personal identifiable information (PII) instances per patient across modalities.



Supplementary Figure 6: Multimodal Anonymizer Pipeline for textual data and audio. The pipeline begins by detecting and redacting PII in filenames using a MLLM, then the files are routed by type and processed with the corresponding processor: tabular data, text documents, and audio files. Audio files are first transcribed to text using whisper-large-v3²⁶, after which all text-based content passes through a regex-based date detection step followed by a date shift (random amount of days within ± 3 years) tool that applies consistent temporal offsets. For tabular data, a dedicated agent identifies and redacts PII-containing columns using a column redaction tool, while all text content is processed by a separate agent that identifies and replaces PII with standardized placeholders. A verification agent then reviews the redacted output (max. 3 times) against the original text to catch any missed identifiers, using the text redaction tool for corrections as well as the date shift tool if it detects an unshifted date. After the redaction for audio files, the text is resynthesized into speech using kokoro-onnx^{27–29} and the redacted text replaced with an acoustic masking signal.



Supplementary Figure 7: Multimodal Anonymizer Pipeline for images, PDFs, videos and DICOM files. After filename-level PII detection, the pipeline routes files by type: DICOM files are split into metadata and image components, videos are processed either frame-by-frame or by first frame only based on user preference, images are passed directly. For DICOM metadata, dates are identified via regex and shifted using the date shift tool, rule-based redaction removes identifiers defined in DICOM guidelines, and an MLLM agent redacts PII in free-text metadata fields. For all modalities, including DICOM, videos, images and PDFs, a temporary PNG representation is generated to be able to send the image to the MLLM. DICOM files and videos then pass through an MLLM agent that determines whether the image represents a volumetric

head scan: if so, a ResNet-UNet model performs defacing (train dataset: size ~1000 examples containing faces and ~15000 Echo and CXR images as negative examples; test dataset: ~300); otherwise, a RetinaNet model performs face detection (train dataset: size ~1400 examples containing faces and ~15000 Echo and CXR images as negative examples; test dataset: size ~3500) and redacts any identified faces. All other modalities are likewise processed through RetinaNet for face detection. EasyOCR²³ then extracts burned-in text with bounding box coordinates from the original data, and an MLLM agent receives both the temporary PNG and the detected text to identify which elements constitute PII. Fuzzy matching aligns the MLLM-identified PII with the OCR bounding boxes to account for recognition errors; if matching fails, which can especially occur with handwritten content, a fine-tuned TrOCR model^{24,25} (train dataset: size ~93000; test dataset: size ~23000) is invoked as a fallback to re-extract the text before reattempting alignment. All redactions are applied to the original file rather than to the temporary PNG representations. A verification agent then reviews the redacted output for residual identifiers, repeating the detection–redaction cycle for up to two iterations. For videos processed in “first-frame-only mode”, redactions identified on the initial frame are propagated to all subsequent frames, and all files are returned in their original format.

Supplementary Table S8. Model Configuration Details.*							
Model Name	Developer	Reasoning Mode	Architecture	Params (total/active)	Access Date	Deployment Type	Max. Context Window
GLM-4.6V ^{33,34}	Z.ai	No Thinking	MoE	106B/12B	04.03.2026	Local	128K
GPT-5.2 ⁴⁸	OpenAI	High Reasoning	MoE	-	22.01.2026	Azure	400K
Kimi-K2.5 (thinking) ^{31,32}	Moonshot AI	Thinking	MoE	1TB/32B	11.02.2026	Local	256K
Kimi-K2.5 (no thinking) ^{31,32}	Moonshot AI	No thinking	MoE	1TB/32B	11.02.2026	Local	256K
Qwen3-VL-4B-Instruct ³⁶	Alibaba Cloud	No thinking	Dense	4B/4B	28.02.2026	Local	256K
Qwen3-VL-8B-Instruct ³⁷	Alibaba Cloud	No thinking	Dense	8B/8B	28.02.2026	Local	256K
Qwen3-VL-30B-A3B-Instruct ³⁸	Alibaba Cloud	No thinking	MoE	30B/3B	24.02.2026	Local	256K
Qwen3-VL-32B-Instruct ³⁹	Alibaba Cloud	No thinking	Dense	32B/32B	25.02.2026	Local	256K
Qwen3-VL-32B-Instruct-AWQ-4bit ⁴²	Alibaba Cloud /cyankiwi	No thinking	Dense (4-bit quantized)	32B/32B	27.02.2026	Local	256K
Qwen3-VL-235B-A22B-Instruct ⁴⁰	Alibaba Cloud	No thinking	MoE	235B/22B	03.03.2026	Local	256K
Qwen3-VL-235B-A22B-Thinking ⁴¹	Alibaba Cloud	Thinking	MoE	235B/22B	09.02.2026	Local	256K
Qwen3.5-9B ⁴⁴	Alibaba Cloud	Thinking	Dense	9B/9B	03.03.2026	Local	262K
Qwen3.5-27B ⁴⁵	Alibaba Cloud	Thinking	Dense	27B/27B	03.03.2026	Local	262K
Qwen3.5-35B-A3B ⁴⁶	Alibaba Cloud	Thinking	MoE	35B/3B	28.02.2026	Local	262K
Qwen3.5-397B-A17B (no thinking) ⁴⁷	Alibaba Cloud	No thinking	MoE	397B/17B	04.03.2026	Local	262K
Qwen3.5-397B-A17B (thinking) ⁴⁷	Alibaba Cloud	Thinking	MoE	397B/17B	04.03.2026	Local	262K

* MoE denotes Mixture-of-Experts.

Supplementary Table S9. Manual reidentification assessment criteria.*

Identifier category	Direct Identifier (one or more identifier were considered high risk for reidentification)	Indirect Identifier (two or more identifiers were considered high risk for reidentification)
Names	Full name (first name + last name); first name only if uncommon; last name only if uncommon, nickname only if uncommon.	Initials; first name; last name; nickname; name abbreviation; masked name.
Geographic subdivisions smaller than a state	Full address; GPS coordinates; geocode.	Street name without number; street number without street; neighborhood; city; county; ZIP code.
Dates directly related to an individual	Age over 89; exact time stamps (level smaller than hours)	Time stamps (level greater than hours).
Telephone numbers	Mobile number; work phone; emergency contact number; voicemail number.	Area code; partial phone number.
Fax numbers	Home fax; employer fax; institutional fax.	Area code; masked fax number.
Email addresses	Personal email; work email; school email; portal email.	Masked email.
Social Security numbers	Full social security number; social security number in any format.	Masked social security number.
Medical record numbers	Medical record number; patient ID; encounter number; accession number if tied to a patient record.	Masked Medical record number or patient ID.
Health plan beneficiary numbers	Insurance member ID; subscriber ID.	Masked member ID or subscriber ID.
Account numbers	Patient financial account; payment account number.	Masked financial account number or payment account number.
Certificate or license numbers	Driver's license; passport number; professional license number.	Masked driver's license, passport number or professional license number.
Vehicle identifiers and serial numbers, including license plates	License plate; vehicle registration number; vehicle serial number.	Masked license plate; vehicle registration number; vehicle serial number.
Device identifiers and serial numbers	Medical device serial number.	Masked medical device serial number.
Web URLs	Patient portal URL; URL containing patient ID; social media account.	Partial URL containing patient ID, masked social media account.
IP addresses	Static home IP; institutional IP.	Truncated IP.
Biometric identifiers	Fingerprint; voiceprint; iris scan; retina scan; facial recognition template; palm print; biometric embedding.	Partial biometric identifiers.

Supplementary Table S9. Manual reidentification assessment criteria.* (continued)		
Identifier category	Direct Identifier (one or more identifier were considered high risk for reidentification)	Indirect Identifier (two or more identifiers were considered high risk for reidentification)
Full-face photographic images and comparable images	Full-face photo; profile face image; video showing face; image where the patient is visually identifiable.	Partial or blurred face, not directly identifiable.
Any other unique identifying number, characteristic, or code	Direct identifier if patient is directly identifiable.	Indirect identifier if patient is not directly identifiable.

**We applied a prespecified, rule-based approach to classify 250 patient records according to reidentification risk. Each record was independently reviewed by two expert reviewers for the presence of direct and indirect identifiers. Records were classified as high risk if they contained at least one direct identifier, such as a full patient name or unique identifying number, or at least two indirect identifiers, such as partial identifiers, specific dates, geographic details, or other contextual elements that could enable linkage. Discrepancies between reviewers were resolved by consensus. The direct/indirect distinction was used as an operational re-identification risk framework in this study.*^{30,60–62}

iaim/multimodal-anonymizer). After initializing the project, the following workflow within the User Interface can be used to perform the anonymization:

1. Model Configuration:

After starting the application, the user selects the LLM to be used for anonymization by providing a model endpoint URL and the model's name. If the model requires an API key, it can be entered in the same form (though locally deployed models typically do not require one). Upon clicking "*Connect to LLM*", the application verifies whether a connection to the specified model can be established. If the connection fails, a descriptive error message is displayed; if successful, the user is directed to the next view, where files can be uploaded for processing.

2. File Upload:

Files and folders can be uploaded via drag and drop or by selecting them through the system's file explorer. Uploaded items can be removed individually or cleared all at once. To modify the model configuration, the user can return to the previous view by clicking on "*LLM Configuration*".

3. Anonymization Configuration:

The user can optionally enable pseudonymization, which saves a CSV file mapping the anonymized filenames to their originals. If a video file has been uploaded, an additional option appears allowing the user to choose whether each frame should be processed individually or whether only the first frame should be anonymized and the redactions applied to all subsequent frames.

4. Prompt Customization (optional):

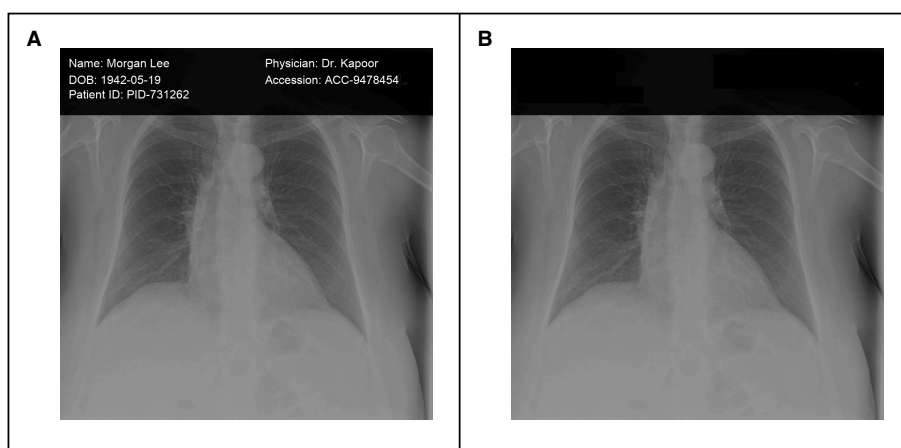
Prompts can be customized by clicking "*Customize Prompts*". The user can define an additional instruction that is applied globally to all prompts sent to the language model during the anonymization process. Beyond this, individual prompts for specific modalities and processing steps can also be adapted. For example, the prompt governing column-level redaction in tabular data or the prompt used during the verification phase.

5. File Processing:

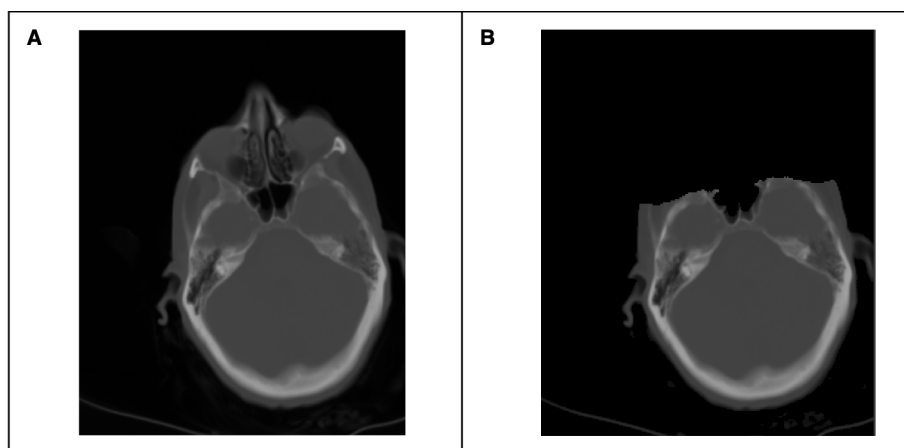
Processing duration primarily depends on the inference speed of the underlying local LLM, making hardware configuration and model size key factors for overall throughput.

6. File Download:

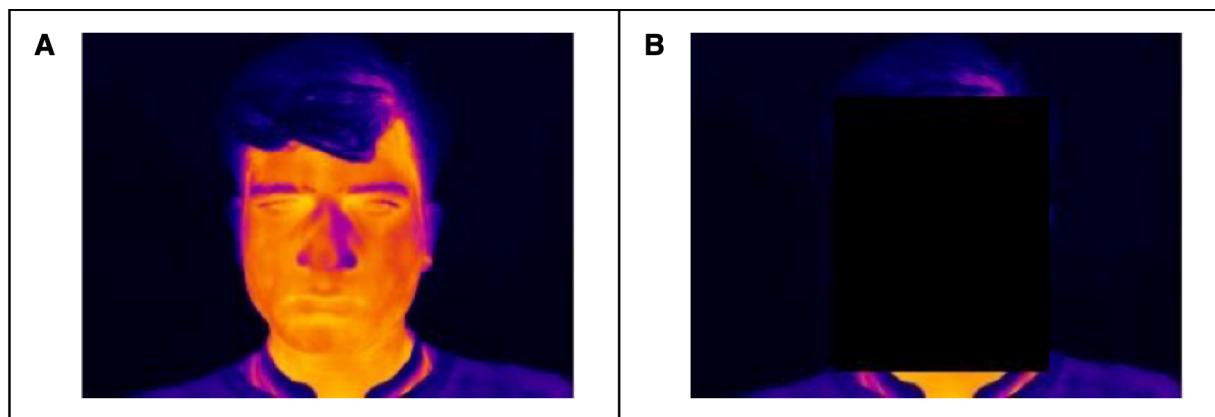
Once all files have been processed, the anonymized results can be downloaded by clicking "*Download Anonymized Files*". When multiple files were processed, the data are provided as a compressed zip archive. If the pseudonymization option was selected, the CSV file mapping original to anonymized filenames is included in the download. To anonymize additional data, the user can click "*Process Another File*" to return to the upload view.



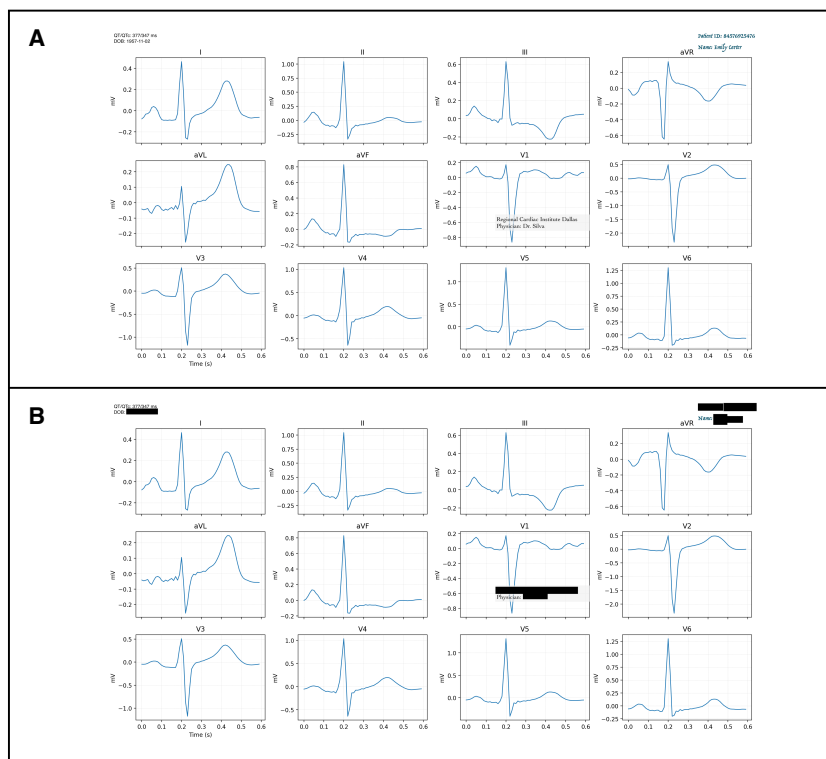
Supplementary Figure 11: Results – Deidentification of a chest X-ray image. (A) Chest X-ray from the RSNA Pneumonia Detection Challenge dataset⁶³ with injected synthetic personally identifiable information. (B) Resulting image of anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).



Supplementary Figure 12: Results – Deidentification of a head CT image. (A) Axial head CT slice from a publicly available dataset¹⁷. (B) Resulting image of anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).



Supplementary Figure 13: Results – Deidentification of Face Image. (A) Image of a person with a face visible (image obtained at Charité – Universitätsmedizin Berlin; written informed consent for publication was obtained from the depicted individual). (B) Resulting image of anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).



Supplementary Figure 14: Results – Deidentification of an ECG. (A) ECG from the PTB-XL dataset⁶⁴ with injected synthetic personally identifiable information. (B) Resulting image after anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).

A Alissa Rieser geb. 01.07.1963, Sehr geehrter Frau Kollegin, sehr geehrter Herr Kollege, wir berichten Ihnen folgend über o. g. Patientin, die sich am 27.10.2023 in unserer teils stationären Behandlung befand. Diagnosen: 1. Pankreas Kopf CA ED 06/2022 2. Diabetes mell II, diät eingestellt 3. V. a. allergische Reaktion gegen Sultamicillin Aktuell Therapie: ERCP mit Stentwechsel am 22.03.2023: Problemlos Einund Vorführen des Gerätes sowie Extraktion der liegenden Prothese, dieses ist incrustiert. Einstellung zur Extraktion ist extrem schwierig. Anschl. Einstellung mit Hilfe des therapeutischen Gerätes gelingt nicht, so dass nach mehreren Versuchen das Gerät gewechselt. Mit analytisches Gerät gelingt das Gallen Gang System zu sondieren und Prothese zu placieren. Danach gutes Abfluß Galle. Beurteilung: Wechsel der Tannenbaum Prothese (11,5 F/8cm in 7,5 F/8 cm), höhergradiger Stenose im Bereich des Duodenum. Hierdurch gelingt Einstellung von Papille nur schwierige. Gutes Abfluß Galle. Patientin bis Heute Abend Nüchter lassen. Epikrise und Therapieempfehlung: Frau Rieser wurde uns zum Routine Stentwechsel bei bek. Pankreas Kopfcarcinom aufgenommen. Der Wechsel wurde am 22.03.2023 komplikationslos durchgeführt. Wir ersuchen Patientin, bis morgen 8.00 Uhr nüchtern zu bleiben; Tee trinken darf sie ab heute um 18.00 Uhr zu sich nehmen. Ihr freundl. Einverst. voraussetzende, empfehlen wir Wiedervorstellung nach entsprechender Termin Vereinbarung (unter 0233-228-312257) zum erneuten Stent Wechsel. Da sich bereits dieses Mal das Stetwechsel als techn. schwierig erwiesen hat, sollte nächstes Wechsel unter stat. Bedingungen erfolgen, da dann möglicherweise ein perkutanes Vorgehen vorteilhaft erwiesen würden. Medikation bei Entlassung: wie gehabt Mit freundlichen kollegialen Grüßen Dra. Ana Lăcrămioara Isărescu	B ***** geb. 05.02.1961, Sehr geehrter Frau Kollegin, sehr geehrter Herr Kollege, wir berichten Ihnen folgend über o. g. Patientin, die sich am 27.10.2020 in unserer teils stationären Behandlung befand. Diagnosen: 1. Pankreas Kopf CA ED 01/2020 2. Diabetes mell II, diät eingestellt 3. V. a. allergische Reaktion gegen Sultamicillin Aktuell Therapie: ERCP mit Stentwechsel am 27.10.2020: Problemlos Einund Vorführen des Gerätes sowie Extraktion der liegenden Prothese, dieses ist incrustiert. Einstellung zur Extraktion ist extrem schwierig. Anschl. Einstellung mit Hilfe des therapeutischen Gerätes gelingt nicht, so dass nach mehreren Versuchen das Gerät gewechselt. Mit analytisches Gerät gelingt das Gallen Gang System zu sondieren und Prothese zu placieren. Danach gutes Abfluß Galle. Beurteilung: Wechsel der Tannenbaum Prothese (11,5 F/8cm in 7,5 F/8 cm), höhergradiger Stenose im Bereich des Duodenum. Hierdurch gelingt Einstellung von Papille nur schwierige. Gutes Abfluß Galle. Patientin bis Heute Abend Nüchter lassen. Epikrise und Therapieempfehlung: ***** wurde uns zum Routine Stentwechsel bei bek. Pankreas Kopfcarcinom aufgenommen. Der Wechsel wurde am 27.10.2020 komplikationslos durchgeführt. Wir ersuchen Patientin, bis morgen 8.00 Uhr nüchtern zu bleiben; Tee trinken darf sie ab heute um 18.00 Uhr zu sich nehmen. Ihr freundl. Einverst. voraussetzende, empfehlen wir Wiedervorstellung nach entsprechender Termin Vereinbarung (unter ***** zum erneuten Stent Wechsel. Da sich bereits dieses Mal das Stetwechsel als techn. schwierig erwiesen hat, sollte nächstes Wechsel unter stat. Bedingungen erfolgen, da dann möglicherweise ein perkutanes Vorgehen vorteilhaft erwiesen würden. Medikation bei Entlassung: wie gehabt Mit freundlichen kollegialen Grüßen *****
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Supplementary Figure 15: Results – Deidentification of a clinical letter. (A) German-language clinical letter from the GRASCCO dataset²² containing personally identifiable information. (B) Resulting image after anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).

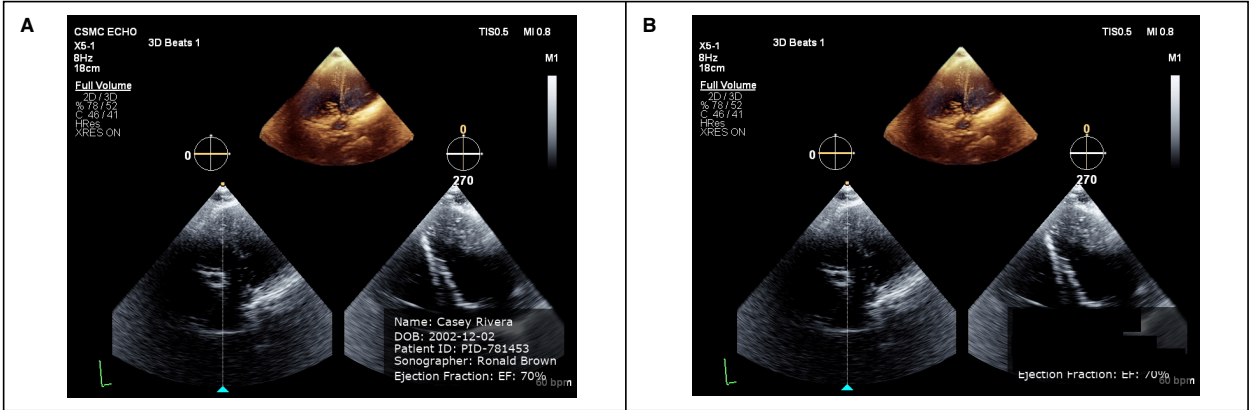
A Name: Marisol E. Hartwell HRN: 00917452 Unit No.: MED-482731 DOB: 08/23/1968 Admission Date: 04/12/2026 Discharge Date: 04/15/2026 Sec F Attending: Dr Naomi Merrilweather Allergies: Hydrocodone (nausea), oxycodone (pruritus) CHIEF COMPLAINT: Abdominal pain and increasing abdominal girth. PRINCIPAL PROCEDURE: 13.04.2026 – Ultrasound-guided therapeutic paracentesis. HISTORY OF PRESENT ILLNESS: Ms. Hartwell is a 57-year-old woman with HIV on antiretroviral therapy, COPD on intermittent 2 L home oxygen, and decompensated hepatitis C cirrhosis complicated by recurrent ascites and prior hepatic encephalopathy. She presented with five days of increasing abdominal distention, early satiety, and diffuse abdominal pressure. She normally undergoes paracentesis every one to two weeks but missed her last appointment because of transportation difficulties. She denied fever, vomiting, gastrointestinal bleeding, dysuria, confusion, or worsening dyspnea. In the ED, temperature was 99.1 F, BP 101/62, HR 106, RR 19, and oxygen saturation 95% on room air. Laboratory studies showed sodium 126, potassium 5.6, creatinine 0.7, AST 218, ALT 121, bilirubin 1.7, INR 1.8, and WBC 7.8. Diagnostic paracentesis showed 186 nucleated cells/mm3 with 14% neutrophils. PAST MEDICAL HISTORY: HCV cirrhosis with recurrent ascites; hepatic encephalopathy; HIV (CD4 228, viral load undetectable 03-18-2026); severe COPD; chronic hyponatremia; bipolar II disorder; PTSD; remote opioid and cocaine use. HOSPITAL COURSE: Decompensated cirrhosis with ascites: Therapeutic paracentesis was performed on 04/13/2026 with removal of 4.1 L of clear yellow fluid. She received 25 g IV albumin. Ascitic fluid cultures showed no growth, and there was no evidence of spontaneous bacterial peritonitis. Her pain and appetite improved significantly. Furosemide 10 mg daily was started; spironolactone was avoided because of hyperkalemia. DISPOSITION: Home with Harbor Light Community Hoapics. FOLLOW-UP: Basic metabolic panel twice weekly. Results to be sent to E. Zhou, MD, Northridge Liver Center, 6175550199. Continue therapeutic paracentesis as needed for comfort. <i>Naomi Merrilweather</i> Naomi Merrilweather, MD 04/15/2026 at 14:37	B Name: [REDACTED] HRN: [REDACTED] Unit No.: [REDACTED] DOB: [REDACTED] Admission Date: [REDACTED] Discharge Date: [REDACTED] Sec F Attending: [REDACTED] Allergies: Hydrocodone (nausea), oxycodone (pruritus) CHIEF COMPLAINT: Abdominal pain and increasing abdominal girth. PRINCIPAL PROCEDURE: [REDACTED] – Ultrasound-guided therapeutic paracentesis. HISTORY OF PRESENT ILLNESS: Ms. Hartwell is a 57-year-old woman with HIV on antiretroviral therapy, COPD on intermittent 2 L home oxygen, and decompensated hepatitis C cirrhosis complicated by recurrent ascites and prior hepatic encephalopathy. She presented with five days of increasing abdominal distention, early satiety, and diffuse abdominal pressure. She normally undergoes paracentesis every one to two weeks but missed her last appointment because of transportation difficulties. She denied fever, vomiting, gastrointestinal bleeding, dysuria, confusion, or worsening dyspnea. In the ED, temperature was 99.1 F, BP 101/62, HR 106, RR 19, and oxygen saturation 95% on room air. Laboratory studies showed sodium 126, potassium 5.6, creatinine 0.7, AST 218, ALT 121, bilirubin 1.7, INR 1.8, and WBC 7.8. Diagnostic paracentesis showed 186 nucleated cells/mm3 with 14% neutrophils. PAST MEDICAL HISTORY: HCV cirrhosis with recurrent ascites; hepatic encephalopathy; HIV (CD4 228, viral load undetectable [REDACTED]); severe COPD; chronic hyponatremia; bipolar II disorder; PTSD; remote opioid and cocaine use. HOSPITAL COURSE: Decompensated cirrhosis with ascites: Therapeutic paracentesis was performed on [REDACTED] with removal of 4.1 L of clear yellow fluid. She received 25 g IV albumin. Ascitic fluid cultures showed no growth, and there was no evidence of spontaneous bacterial peritonitis. Her pain and appetite improved significantly. Furosemide 10 mg daily was started; spironolactone was avoided because of hyperkalemia. DISPOSITION: [REDACTED] FOLLOW-UP: Basic metabolic panel twice weekly. Results to be sent to [REDACTED]. Continue therapeutic paracentesis as needed for comfort. [REDACTED] MD [REDACTED] at 14:37
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Supplementary Figure 16: Results – Deidentification of a clinical discharge summary. (A) Fully synthetic clinical discharge summary with synthetic personally identifiable information, including handwritten annotations. (B) Resulting image after anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).

A								
record_uid	pid	report_kind	performed_at	verified_at	accession	procedure_title	facility_name	report_body
RAD-2025-000731	PT-730421	Diagnostic Radiology	2025-04-18 09:10:00	2025-04-18 10:02:00	25-0418-7732	CHEST RADIOGRAPH; TWO VIEWS	Cedar Ridge Imaging Center	PT-730421, Emily Carter, 2025-04-18 ORDERING CLINICIAN: Dr. Lila Renn EXAMINATION: CHEST RADIOGRAPHS, PA AND LATERAL. INDICATION: 48-year-old woman with three days of cough and mild shortness of breath. COMPARISON: Chest radiograph dated 2024-12-03. The cardiomedialastinal silhouette is within normal limits. Mild linear opacity at the left lung base is compatible with subsegmental atelectatic change. No focal air-space consolidation, pleural effusion, or pneumothorax is identified. No acute osseous abnormality is seen. Findings were discussed by phone with Lara Smith, md (+44823746222) Mild left basilar linear atelectatic change. No focal pneumonia.
RAD-2025-001284	PT-895502	Ultrasound	2025-07-22 14:35:00	2025-07-22 15:26:00	25-0722-4189	COMPLETE ABDOMINAL ULTRASOUND	Stonebridge Diagnostic Clinic	ORDERING CLINICIAN: Omar Venn EXAMINATION: COMPLETE ABDOMINAL ULTRASOUND on 07/22/2025 INDICATION: 35-year-old man with intermittent right upper quadrant discomfort. The liver is normal in size and echogenicity without focal lesion. The main portal vein is patent with hepatopetal flow. No intrahepatic biliary ductal dilation is present. The common bile duct measures 4 mm. The gallbladder contains no stones and shows no wall thickening or pericholecystic fluid. Dr Michael Harris was paged IMPRESSION: Unremarkable complete abdominal ultrasound.

B								
record_uid	pid	report_kind	performed_at	verified_at	accession	procedure_title	facility_name	report_body
*****	*****	Diagnostic Radiology	2023-08-07 09:10:00	2023-08-07 10:02:00	*****	CHEST RADIOGRAPH; TWO VIEWS	*****	***** 2023-08-07 ORDERING CLINICIAN: ***** EXAMINATION: CHEST RADIOGRAPHS, PA AND LATERAL. INDICATION: 48-year-old woman with three days of cough and mild shortness of breath. COMPARISON: Chest radiograph dated 2023-03-24. The cardiomedialastinal silhouette is within normal limits. Mild linear opacity at the left lung base is compatible with subsegmental atelectatic change. No focal air-space consolidation, pleural effusion, or pneumothorax is identified. No acute osseous abnormality is seen. Findings were discussed by phone with ***** (*****) Mild left basilar linear atelectatic change. No focal pneumonia.
*****	*****	Ultrasound	2023-11-10 14:35:00	2023-11-10 15:26:00	*****	COMPLETE ABDOMINAL ULTRASOUND	*****	ORDERING CLINICIAN: ***** EXAMINATION: COMPLETE ABDOMINAL ULTRASOUND on 11/10/2023 INDICATION: 35-year-old man with intermittent right upper quadrant discomfort. The liver is normal in size and echogenicity without focal lesion. The main portal vein is patent with hepatopetal flow. No intrahepatic biliary ductal dilation is present. The common bile duct measures 4 mm. The gallbladder contains no stones and shows no wall thickening or pericholecystic fluid. ***** was paged IMPRESSION: Unremarkable complete abdominal ultrasound.

Supplementary Figure 17: Results – Deidentification of a table. (A) Fully synthetic clinical table containing synthetic personally identifiable information across multiple fields. (B) Resulting table after anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).



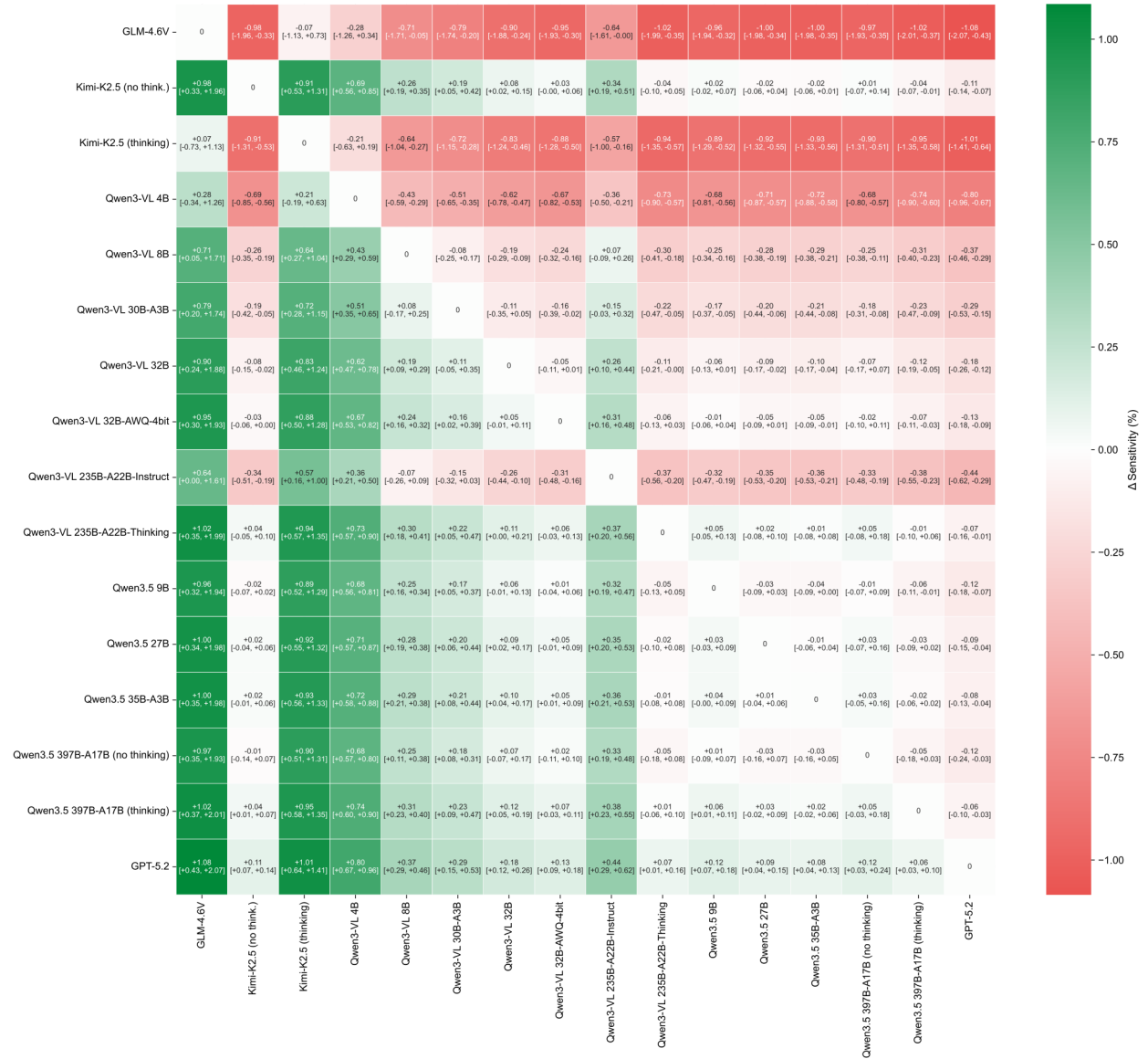
Supplementary Figure 18: Results – Deidentification of an echocardiogram. (A) Echocardiographic from the 3D echocardiography dataset accompanying EchoSlicer⁶⁵ with injected synthetic personally identifiable information in the DICOM. (B) Resulting image after anonymization with the Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking).

Supplementary Table S19. Deidentification Sensitivity, Critical Clinical Preservation and Clinical Preservation.

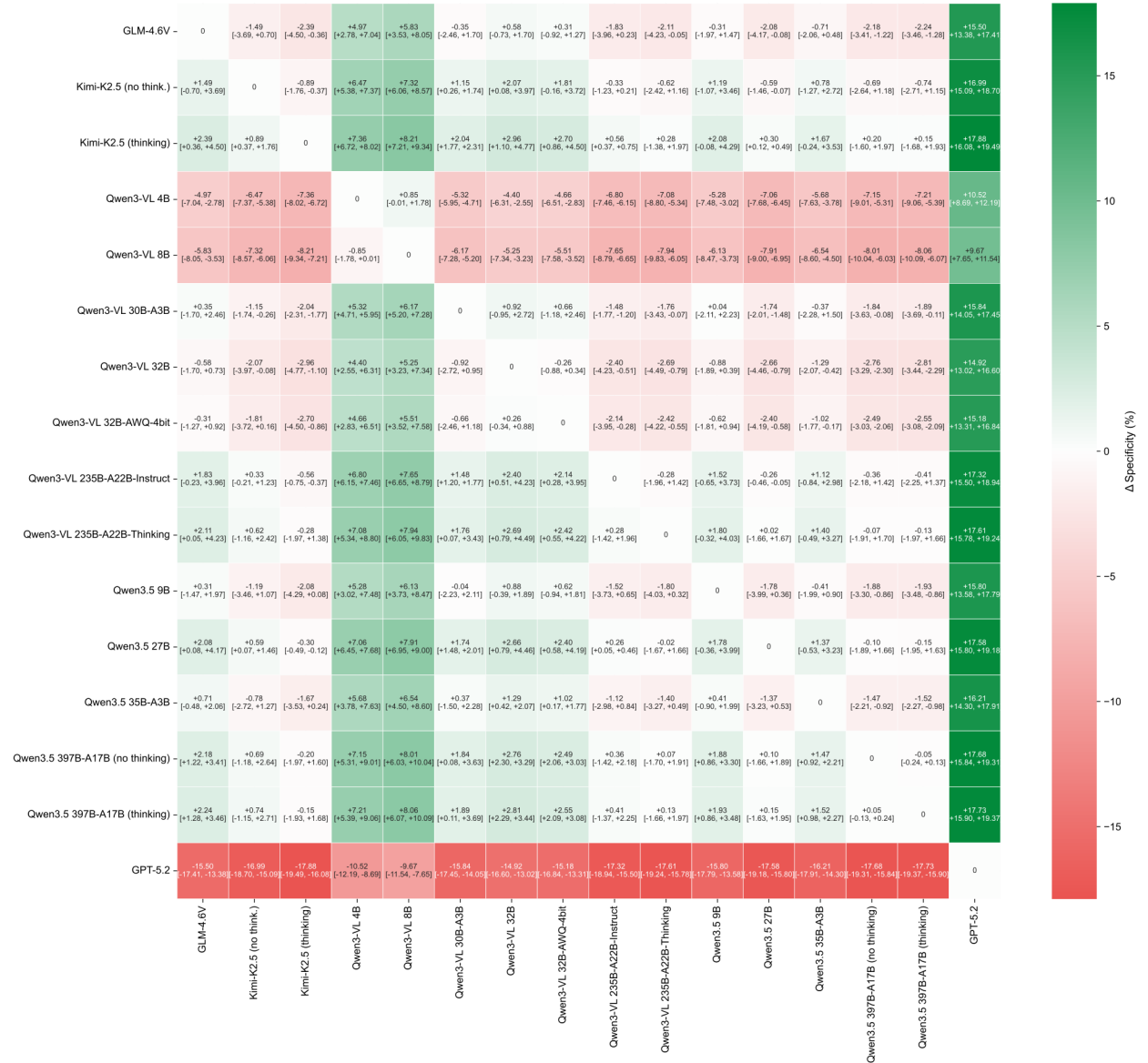
Modality and Metric ($\pm$ 95% CI)	Model
	Multimodal Anonymizer (Qwen3-VL-235B-A22B-Thinking)
Overall (per-patient)	
Deidentification Sensitivity	98.80 $\pm$ (97.20, 100)
Critical Clinical Preservation	99.60 $\pm$ (98.80, 100)
Overall	
Deidentification Sensitivity (per-P11)	99.82 $\pm$ (99.76, 99.88)
Clinical Preservation (per-file)	99.61 $\pm$ (99.51, 99.71)
Images	
Deidentification Sensitivity (per-P11)	99.71 $\pm$ (99.46, 99.90)
Clinical Preservation (per-file)	99.28 $\pm$ (98.81, 99.68)
3D Images	
Deidentification Sensitivity (per-P11)	100
Clinical Preservation (per-file)	99.40 $\pm$ (98.60, 100)
Face Images	
Deidentification Sensitivity (per-P11)	100
Clinical Preservation (per-file)	100
PDF	
Deidentification Sensitivity (per-P11)	99.23 $\pm$ (98.78, 99.63)
Clinical Preservation (per-file)	98.71 $\pm$ (98.19, 99.18)
Text	
Deidentification Sensitivity (per-P11)	99.47 $\pm$ (99.01, 99.83)
Clinical Preservation (per-file)	98.02 $\pm$ (97.19, 98.77)
Text (German)	
Deidentification Sensitivity (per-P11)	99.94 $\pm$ (99.84, 100)
Clinical Preservation (per-file)	100
Text (Handwritten)	
Deidentification Sensitivity (per-P11)	99.08 $\pm$ (98.30, 99.70)
Clinical Preservation (per-file)	99.90 $\pm$ (99.70, 100)
Tables	
Deidentification Sensitivity (per-P11)	99.95 $\pm$ (99.89, 100)
Clinical Preservation (per-file)	99.96 $\pm$ (99.89, 100)
Metadata	
Deidentification Sensitivity (per-P11)	100
Clinical Preservation (per-file)	100
Audio	
Deidentification Sensitivity (per-P11)	100
Clinical Preservation (per-file)	99.80 $\pm$ (99.39, 100)

Supplementary Table S20. Performance Metrics of all tested Models. (continued)

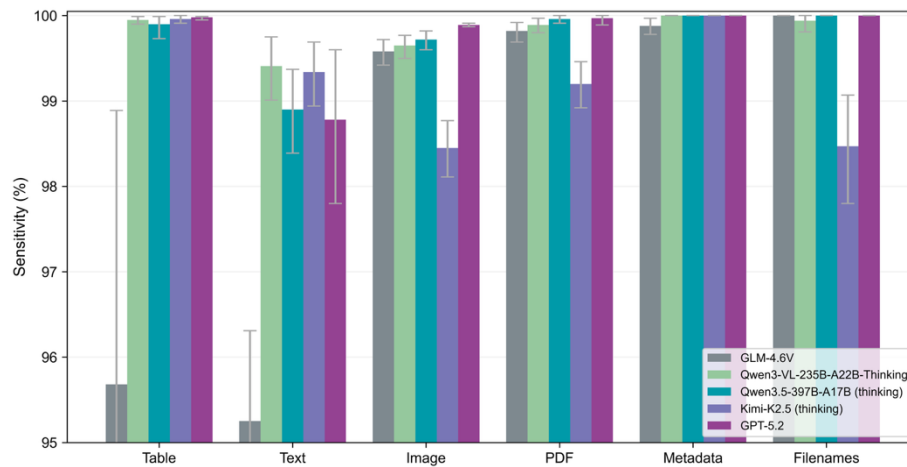
Modality and Metric (± 95% CI)	Model														
	GPT-5.2	Kimi-K2.5 (thinking)	Kimi-K2.5 (no thinking)	Qwen3-VL-235B-A22B-Thinking	Qwen3-VL-235B-A22B-Instruct	Qwen3-VL-32B-Instruct	Qwen3-VL-32B-AWQ-4bit	Qwen3-VL-30B-A3B-Instruct	Qwen3-VL-4B-Instruct	Qwen3.5-397B-A17B (no thinking)	Qwen3.5-397B-A17B (thinking)	Qwen3.5-35B-A3B	Qwen3.5-27B	Qwen3.5-9B	GLM-4.6V
PDF															
Sensitivity	99.97 ± (99.89, 100)	99.20 ± (98.92, 99.46)	99.11 ± (98.80, 99.39)	99.89 ± (99.80, 99.97)	99.71 ± (99.54, 99.86)	99.89 ± (99.81, 99.97)	99.89 ± (99.81, 99.97)	99.68 ± (99.51, 99.83)	93.09 ± (92.21, 93.92)	99.95 ± (99.89, 99.99)	99.96 ± (99.91, 100)	99.95 ± (99.90, 99.99)	99.94 ± (99.88, 99.99)	99.92 ± (99.84, 99.98)	99.82 ± (99.69, 99.92)
Specificity	95.51 ± (95.41, 95.62)	95.72 ± (95.66, 95.78)	95.68 ± (95.62, 95.74)	95.72 ± (95.67, 95.77)	95.72 ± (95.66, 95.77)	95.78 ± (95.72, 95.83)	95.78 ± (95.72, 95.83)	95.50 ± (95.43, 95.57)	96.05 ± (95.96, 96.13)	95.78 ± (95.73, 95.83)	95.78 ± (95.73, 95.83)	95.77 ± (95.71, 95.83)	95.81 ± (95.76, 95.86)	95.79 ± (95.73, 95.84)	95.76 ± (95.71, 95.82)
FNR (per-token)	0.05 ± (0.00, 0.17)	1.21 ± (0.80, 1.65)	1.40 ± (0.94, 1.92)	0.21 ± (0.04, 0.45)	0.50 ± (0.24, 0.82)	0.14 ± (0.04, 0.26)	0.14 ± (0.04, 0.27)	0.55 ± (0.28, 0.86)	8.66 ± (7.58, 9.75)	0.08 ± (0.01, 0.19)	0.05 ± (0.00, 0.11)	0.05 ± (0.01, 0.12)	0.07 ± (0.02, 0.13)	0.10 ± (0.02, 0.19)	0.30 ± (0.13, 0.51)
FPR (per-token)	4.49 ± (4.38, 4.59)	4.28 ± (4.22, 4.34)	4.32 ± (4.26, 4.38)	4.28 ± (4.22, 4.33)	4.28 ± (4.23, 4.34)	4.22 ± (4.17, 4.28)	4.22 ± (4.17, 4.27)	4.50 ± (4.43, 4.57)	3.95 ± (3.87, 4.04)	4.21 ± (4.16, 4.27)	4.22 ± (4.17, 4.27)	4.23 ± (4.17, 4.29)	4.19 ± (4.14, 4.24)	4.21 ± (4.16, 4.27)	4.24 ± (4.18, 4.29)
Metadata															
Sensitivity	100	100	100	100	100	100	100	100	99.61 ± (99.45, 99.76)	99.76 ± (99.28, 100)	100	100	99.78 ± (99.33, 100)	100	99.88 ± (99.78, 99.97)
Specificity	97.16 ± (96.42, 97.89)	98.20 ± (98.02, 98.34)	98.39 ± (98.39, 98.40)	98.39 ± (98.39, 98.39)	98.39 ± (98.39, 98.39)	98.39 ± (98.39, 98.39)	98.39 ± (98.39, 98.39)	98.39 ± (98.39, 98.39)	98.64 ± (98.58, 98.69)	97.93 ± (97.64, 98.20)	97.97 ± (97.69, 98.21)	98.39 ± (98.39, 98.39)	98.39 ± (98.39, 98.39)	97.65 ± (97.28, 97.97)	98.25 ± (98.07, 98.39)
FNR (per-token)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.39 ± (0.24, 0.55)	0.24 ± (0.00, 0.72)	0.00	0.00	0.22 ± (0.00, 0.67)	0.00	0.12 ± (0.03, 0.22)
FPR (per-token)	2.03 ± (1.59, 2.52)	1.33 ± (1.24, 1.43)	1.25 ± (1.24, 1.27)	1.24 ± (1.22, 1.26)	1.22 ± (1.20, 1.24)	1.24 ± (1.22, 1.26)	1.23 ± (1.21, 1.25)	1.25 ± (1.24, 1.27)	1.20 ± (1.17, 1.24)	1.48 ± (1.35, 1.63)	1.44 ± (1.32, 1.59)	1.24 ± (1.22, 1.25)	1.26 ± (1.24, 1.28)	1.69 ± (1.49, 1.91)	1.34 ± (1.25, 1.45)
Filenames															
Sensitivity	100	98.47 ± (97.80, 99.07)	100	99.94 ± (99.81, 100)	100	99.89 ± (99.79, 99.98)	100	100	99.97 ± (99.92, 100)	100	100	100	100	100	100
Specificity	98.31 ± (98.22, 98.40)	99.71 ± (99.56, 99.85)	97.51 ± (97.23, 97.77)	98.81 ± (98.45, 99.15)	99.26 ± (99.07, 99.43)	98.19 ± (97.96, 98.40)	98.20 ± (97.98, 98.41)	99.43 ± (99.24, 99.58)	99.58 ± (99.42, 99.73)	100	100	99.98 ± (99.96, 100)	99.83 ± (99.68, 99.95)	100	99.86 ± (99.79, 99.92)
FNR (per-token)	0.00	1.53 ± (0.93, 2.20)	0.00	0.06 ± (0.00, 0.19)	0.00	0.11 ± (0.02, 0.21)	0.00	0.00	0.03 ± (0.00, 0.08)	0.00	0.00	0.00	0.00	0.00	0.00
FPR (per-token)	1.69 ± (1.60, 1.78)	0.29 ± (0.15, 0.44)	2.49 ± (2.23, 2.77)	1.19 ± (0.86, 1.56)	0.74 ± (0.57, 0.93)	1.81 ± (1.60, 2.04)	1.80 ± (1.59, 2.02)	0.57 ± (0.42, 0.76)	0.42 ± (0.27, 0.58)	0.00	0.00	0.02 ± (0.00, 0.04)	0.17 ± (0.05, 0.32)	0.00	0.14 ± (0.08, 0.21)
Date Shift															
Shift Rate	98.66 ± (98.42, 98.86)	99.64 ± (99.42, 99.81)	99.52 ± (99.27, 99.72)	99.40 ± (98.81, 99.78)	99.57 ± (99.34, 99.76)	94.21 ± (91.35, 96.65)	92.77 ± (90.23, 95.05)	87.44 ± (84.23, 90.48)	59.22 ± (54.52, 63.95)	99.43 ± (98.92, 99.77)	99.63 ± (99.41, 99.80)	99.37 ± (98.98, 99.67)	99.59 ± (99.37, 99.77)	96.46 ± (94.65, 97.94)	98.04 ± (96.94, 98.94)
Redaction Rate	1.21 ± (1.02, 1.45)	0.09 ± (0.02, 0.18)	0.15 ± (0.05, 0.28)	0.27 ± (0.00, 0.80)	0.09 ± (0.01, 0.19)	5.36 ± (2.93, 8.24)	5.23 ± (3.16, 7.48)	12.26 ± (9.19, 15.35)	34.82 ± (28.74, 39.43)	0.25 ± (0.02, 0.69)	0.05 ± (0.01, 0.12)	0.07 ± (0.02, 0.15)	0.20 ± (0.09, 0.32)	2.86 ± (1.47, 4.56)	1.30 ± (0.56, 2.25)
FNR (per-token)	0.02 ± (0.01, 0.03)	0.00 ± (0.00, 0.01)	0.00 ± (0.00, 0.01)	0.00 ± (0.00, 0.01)	0.00 ± (0.00, 0.01)	0.01 ± (0.00, 0.01)	0.06 ± (0.03, 0.11)	0.00 ± (0.00, 0.01)	0.20 ± (0.15, 0.27)	0.00 ± (0.00, 0.01)	0.00 ± (0.00, 0.01)	0.01 ± (0.00, 0.02)	0.00 ± (0.00, 0.03)	0.01 ± (0.00, 0.04)	0.01 ± (0.00, 0.04)



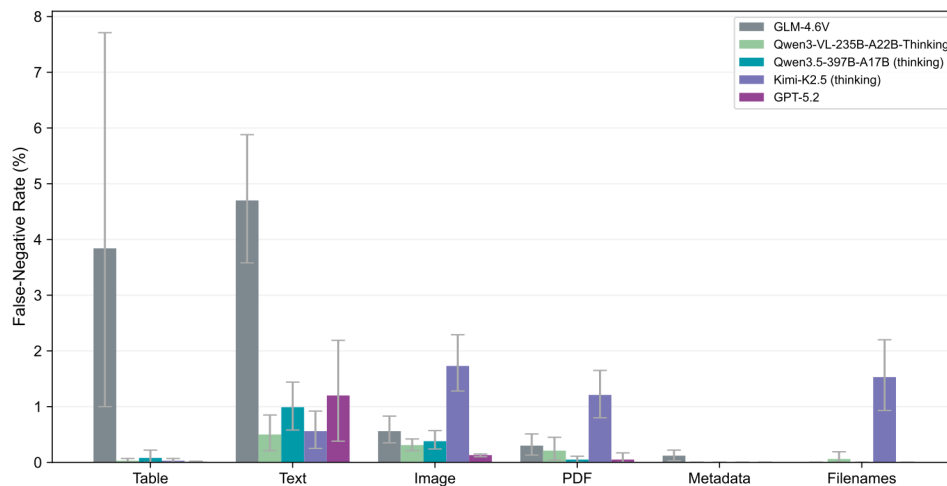
Supplementary Figure 21: Pairwise bootstrap comparison of deidentification sensitivity (per-PII) across all tested models. Each cell displays the mean difference between the deidentification sensitivities (Δ Sensitivity) of the row model and the column model. Values in brackets denote the 95% confidence intervals.



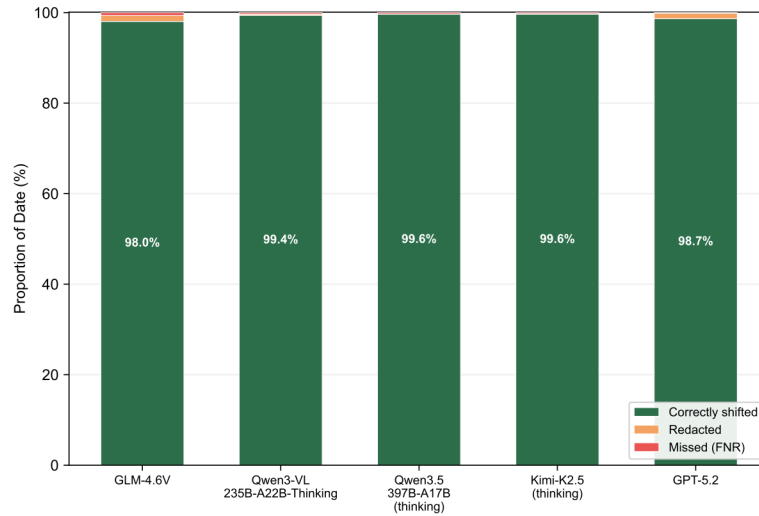
Supplementary Figure 22: Pairwise bootstrap comparison of deidentification specificity (per-PII) across all tested models. Each cell displays the mean difference between specificities (Δ Specificity) of the row model and the column model. Values in brackets denote the 95% confidence intervals.



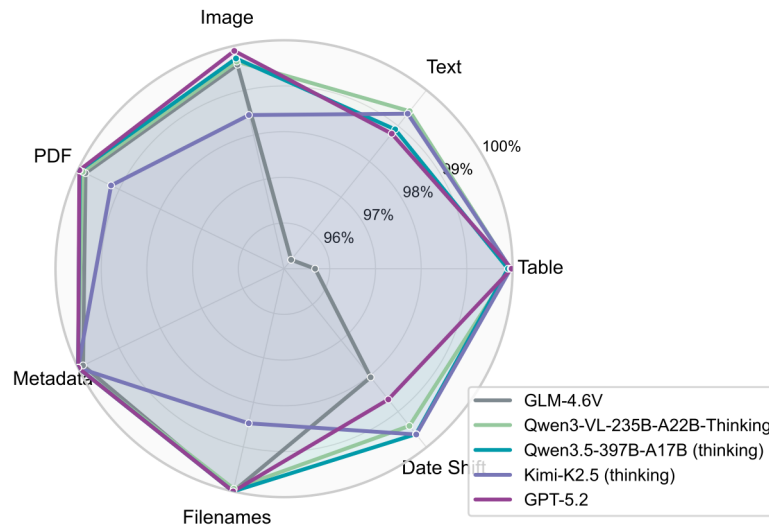
Supplementary Figure 23: Per modality deidentification sensitivity (per-PII) of the five largest evaluated models. Bar charts show deidentification sensitivity for the five largest evaluated models, each with 95% confidence intervals.



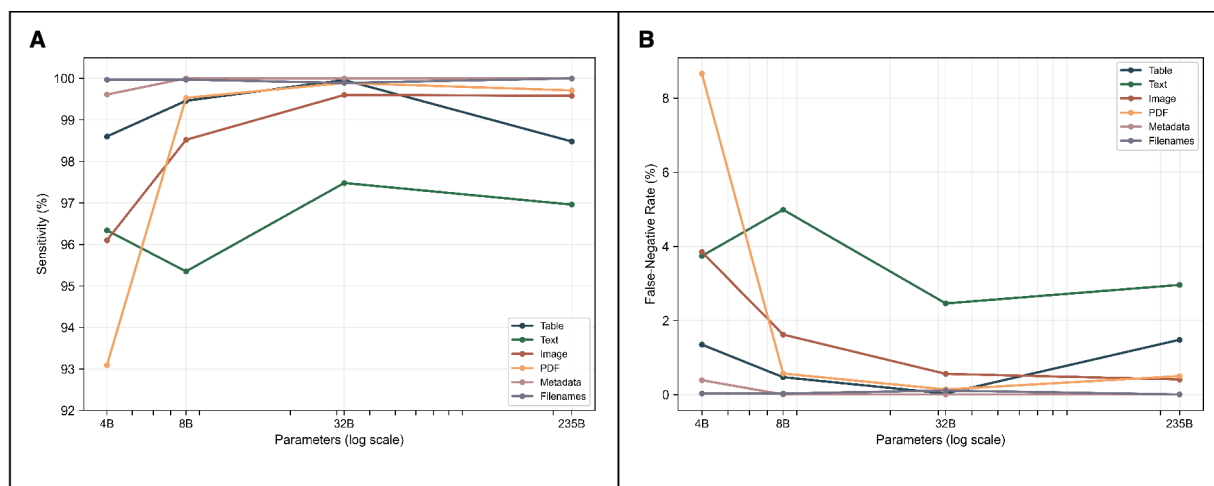
Supplementary Figure 24: Per modality deidentification false-negative rates of the five largest evaluated models. Bar charts display the deidentification per-token false-negative rate per modality for the five largest evaluated models with 95% confidence intervals.



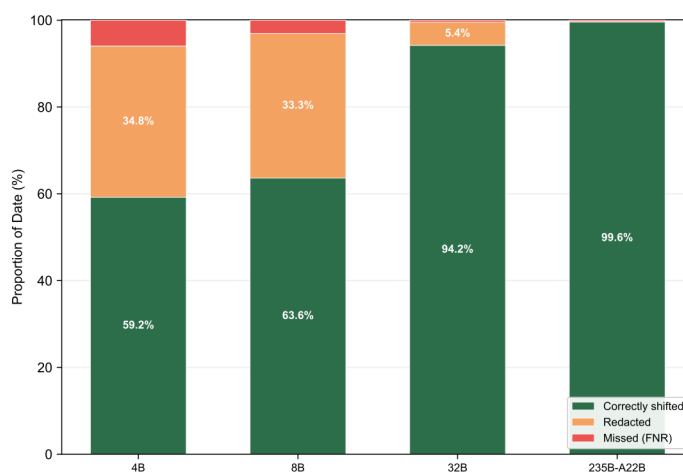
Supplementary Figure 25: Date shifting performance of the five largest evaluated models. Stacked bar charts display the proportion of date entities that were correctly shifted, redacted, or missed across the five largest evaluated models.



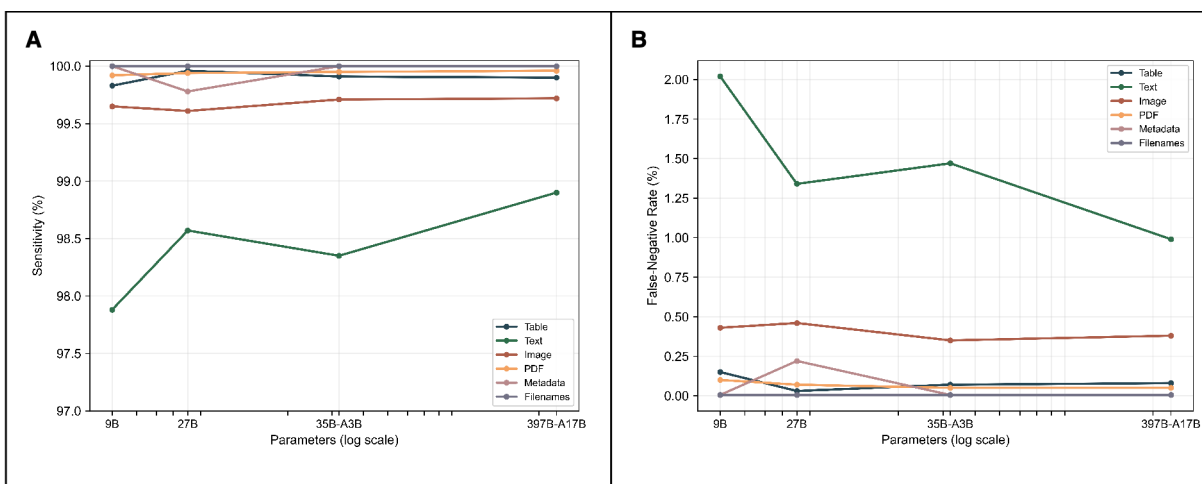
Supplementary Figure 26: Per modality deidentification sensitivity profiles of the five largest evaluated models. Per-PII deidentification sensitivity across document element types and date shift (shift rate is displayed) for the five largest evaluated models.



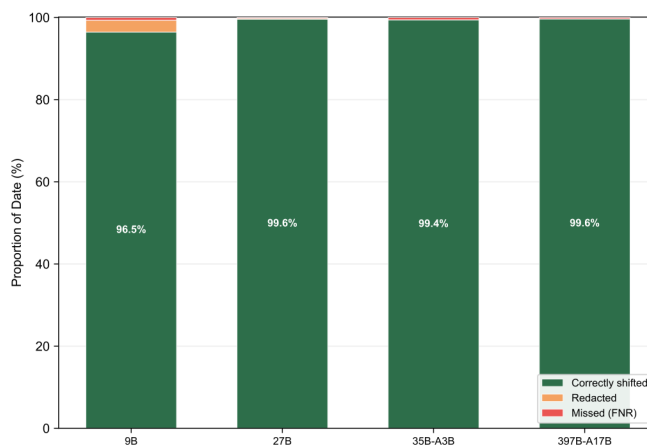
Supplementary Figure 27: Per modality deidentification sensitivity and false-negative rate across Qwen3-VL-Instruct model scales. (Panel A) Deidentification sensitivity (per-PII) and, (Panel B) false-negative rate (per-token) across Qwen3-VL-Instruct model sizes (4B to 235B parameters), stratified by modality.



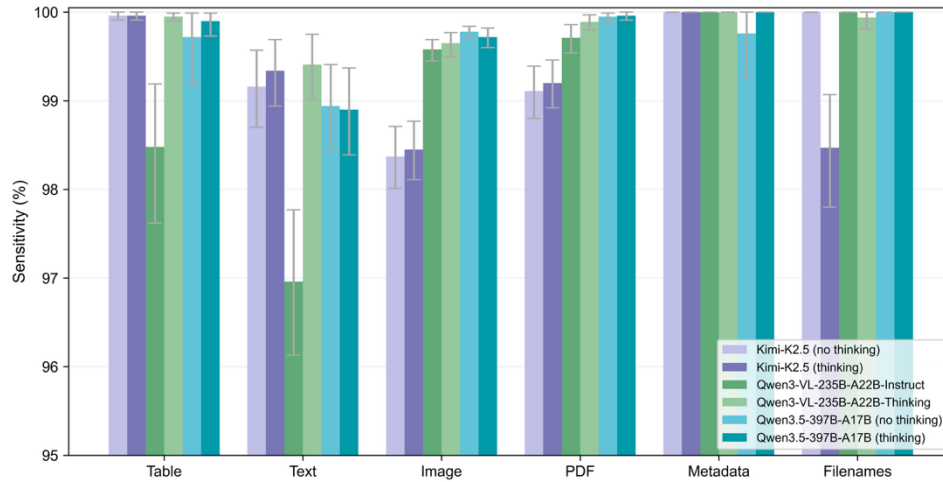
Supplementary Figure 28: Date shifting performance across Qwen3-VL-Instruct model sizes. Stacked bar charts display the proportion of date entities that were correctly shifted, redacted, or missed (false-negative rate, FNR) for four Qwen3-VL-Instruct model sizes (4B to 235B parameters).



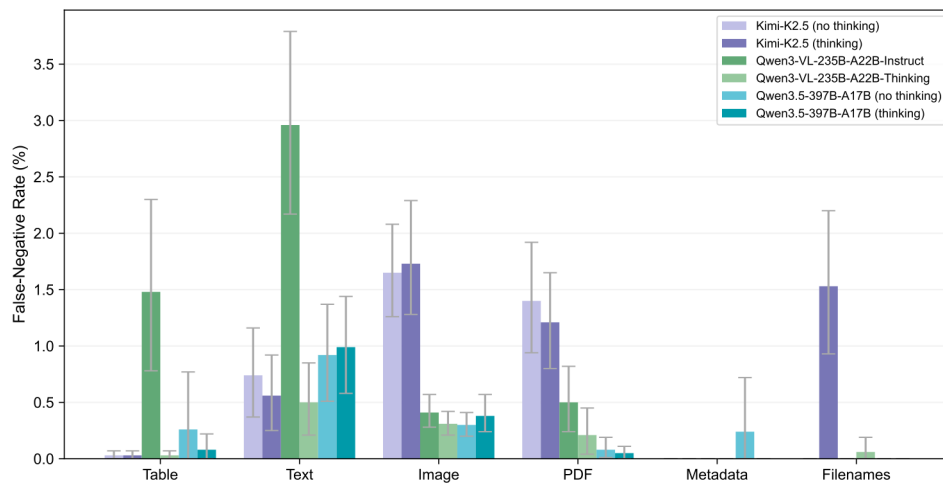
Supplementary Figure 29: Per modality deidentification sensitivity and false-negative rate across Qwen3.5 model sizes. Per-PII deidentification sensitivity (Panel A) and per-token false-negative rate (FNR) (Panel B) across Qwen3.5 model sizes (9B to 397B-A17B parameters), stratified by modality.



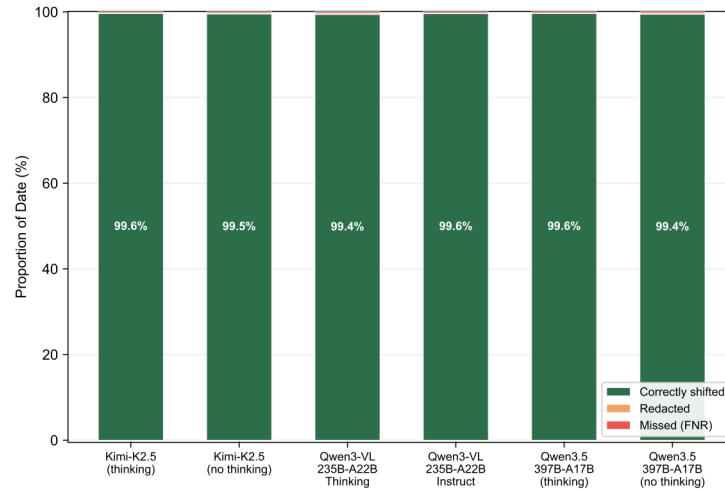
Supplementary Figure 30: Date shifting performance across Qwen3.5 model sizes. Stacked bar charts display the proportion of date entities that were correctly shifted, redacted, or missed (false-negative rate, FNR) for four Qwen3.5 model sizes.



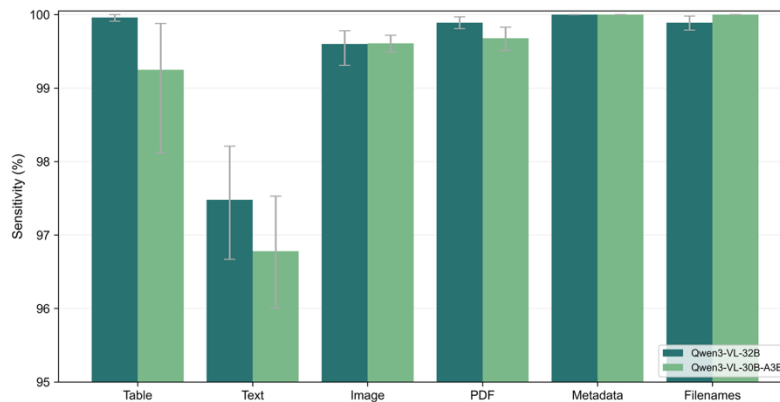
Supplementary Figure 31: Impact of thinking mode on per modality deidentification sensitivity across model variants. Bar charts compare per-PII deidentification sensitivity between thinking and non-thinking variants of Kimi-K2.5, Qwen3-VL-235B-A22B, and Qwen3.5-397B-A17B, with 95% confidence intervals.



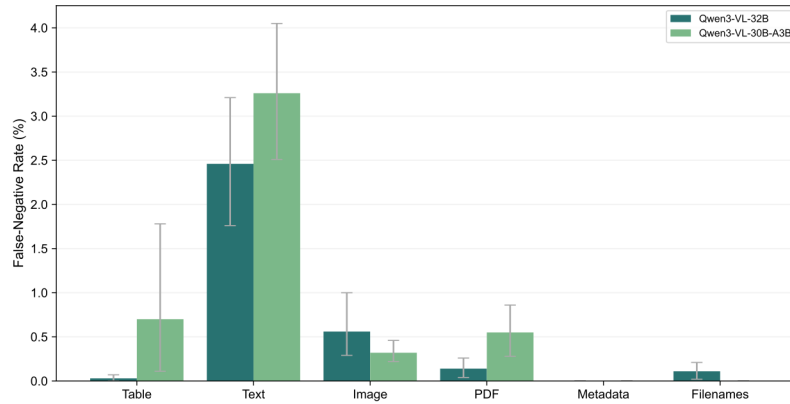
Supplementary Figure 32: Impact of thinking mode on per modality false-negative rates across model variants. Bar charts compare per-token false-negative between thinking and non-thinking variants of Kimi-K2.5, Qwen3-VL-235B-A22B, and Qwen3.5-397B-A17B, with 95% confidence intervals.



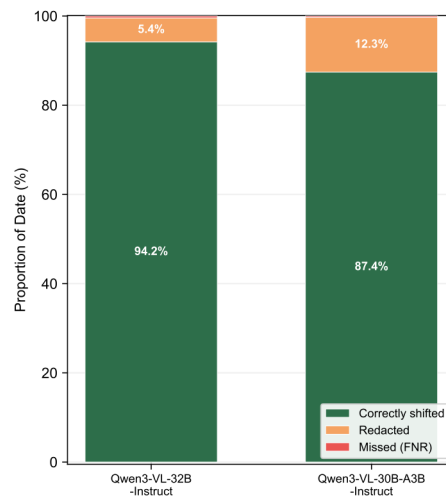
Supplementary Figure 33: Date shifting performance comparison between thinking and non-thinking model variants. Stacked bar charts display the proportion of date entities that were correctly shifted, redacted, or missed (false-negative rate, FNR) for three model families, each evaluated with and without thinking enabled.



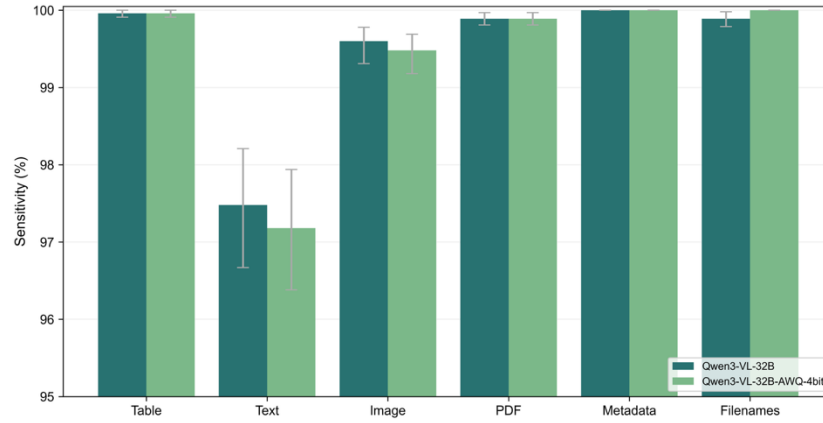
Supplementary Figure 34: Comparison of per modality deidentification sensitivity between a dense and a mixture-of-experts model architecture. Bar charts compare per-PII deidentification sensitivity between Qwen3-VL-32B-Instruct (dense) and Qwen3-VL-30B-A3B-Instruct (mixture-of-experts), with 95% confidence intervals.



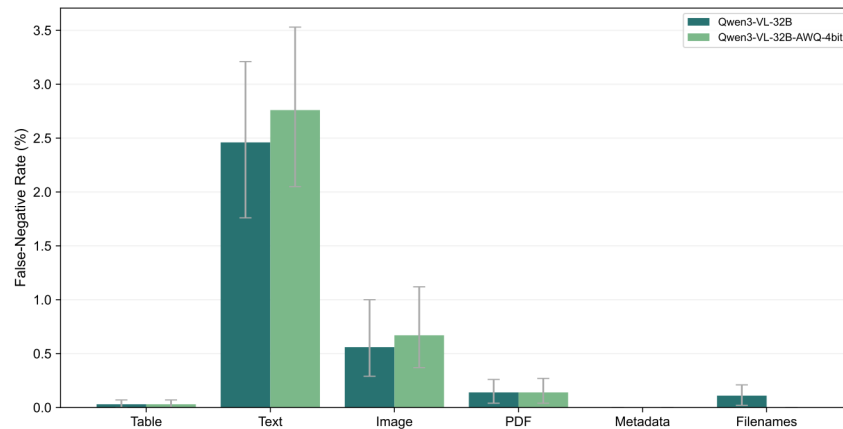
Supplementary Figure 35: Comparison of per modality deidentification false-negative rates between a dense and a mixture-of-experts model architecture. Bar charts compare deidentification per-token false-negative rates between Qwen3-VL-32B-Instruct (dense) and Qwen3-VL-30B-A3B-Instruct (mixture-of-experts), with 95% confidence intervals.



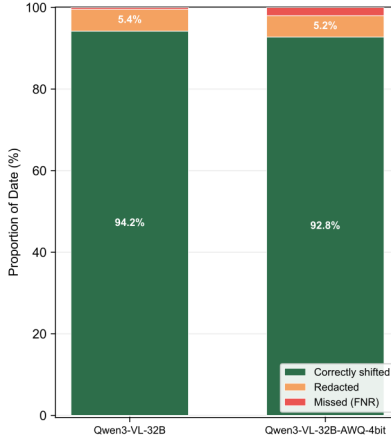
Supplementary Figure 36: Date shifting performance comparison between dense and mixture-of-experts model architectures. Stacked bar charts display the proportion of date entities that were correctly shifted, redacted, or missed (false-negative rate, FNR) for two similarly sized Qwen3-VL-Instruct models with different architectures.



Supplementary Figure 37: Effect of quantization on per modality deidentification sensitivity. Bar charts compare per-PII deidentification sensitivity between Qwen3-VL-32B-Instruct (full precision) and its AWQ 4-bit quantized variant, with 95% confidence intervals.



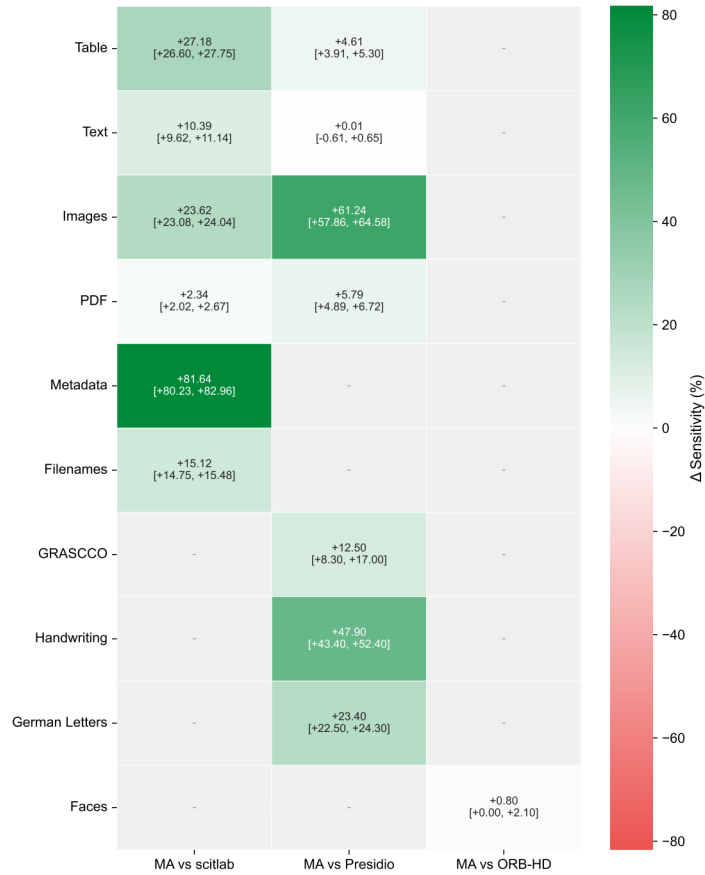
Supplementary Figure 38: Effect of quantization on per modality deidentification false-negative rates. Bar charts compare per-token false-negative rates between Qwen3-VL-32B-Instruct (full precision) and its AWQ 4-bit quantized variant, with 95% confidence intervals.



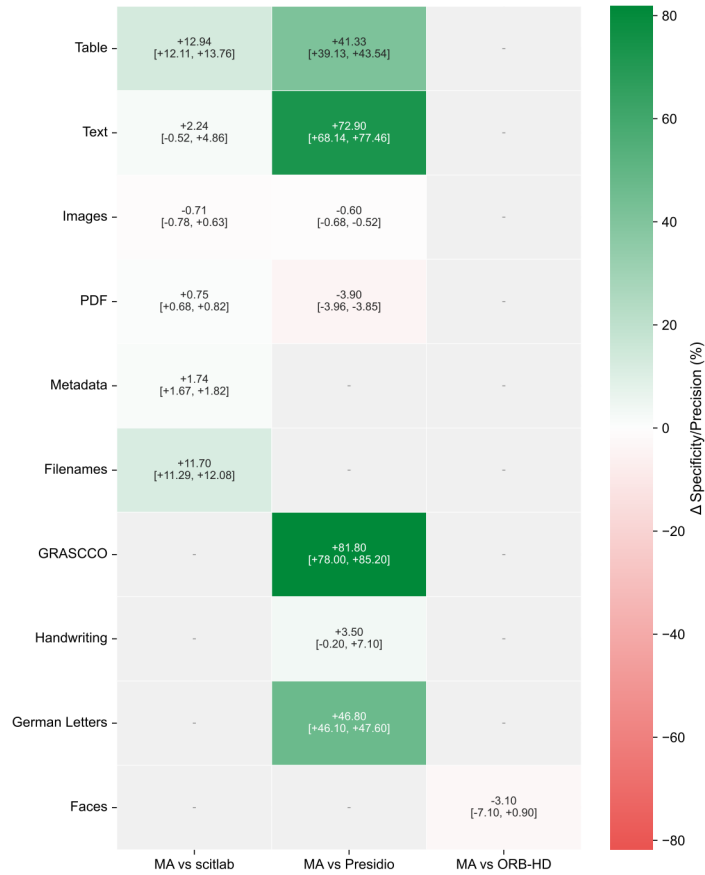
Supplementary Figure 39: Date shifting performance comparison between quantized and non-quantized model variants. Stacked bar charts display the proportion of date entities that were correctly shifted, redacted, or missed (false-negative rate, FNR) for the Qwen3-VL-32B model in its original and AWQ-4bit quantized form.

Supplementary Table S40. Reproducibility Experiment Results.*			
Modality and Metric ($\pm$ 95% CI)	Model	Modality and Metric ($\pm$ 95% CI)	Model
	Kimi-K2.5 (no thinking)		Kimi-K2.5 (no thinking)
Table		PDF	
Sensitivity	99.96 $\pm$ (99.17, 99.96)	Sensitivity	99.88 $\pm$ (99.82, 99.93)
Specificity	80.74 $\pm$ (80.68, 80.79)	Specificity	95.76 $\pm$ (95.75, 95.76)
FNR (per-token)	0.31 $\pm$ (0.04, 0.83)	FNR (per-token)	0.19 $\pm$ (0.13, 0.27)
FPR (per-token)	8.11 $\pm$ (8.08, 8.13)	FPR (per-token)	4.24 $\pm$ (4.24, 4.25)
Text		Metadata	
Sensitivity	98.77 $\pm$ (98.11, 99.11)	Sensitivity	99.67 $\pm$ (99.02, 100)
Specificity	96.50 $\pm$ (96.45, 96.55)	Specificity	98.39 $\pm$ (98.39, 98.39)
FNR (per-token)	1.23 $\pm$ (0.89, 1.89)	FNR (per-token)	0.33 $\pm$ (0.00, 0.98)
FPR (per-token)	2.74 $\pm$ (2.69, 2.79)	FPR (per-token)	1.24 $\pm$ (1.24, 1.24)
Image		Filenames	
Sensitivity	99.63 $\pm$ (99.31, 99.81)	Sensitivity	100
Specificity	99.22 $\pm$ (99.21, 99.23)	Specificity	97.52 $\pm$ (97.51, 97.54)
FNR (per-token)	0.44 $\pm$ (0.26, 0.75)	FNR (per-token)	0.00
FPR (per-token)	0.78 $\pm$ (0.77, 0.79)	FPR (per-token)	2.48 $\pm$ (2.46, 2.49)
		Date Shift	
		Shift Rate	99.51 $\pm$ (99.48, 99.53)
		Redaction Rate	0.16 $\pm$ (0.14, 0.19)
		FNR (per-token)	0.00

* 100 repeated experiments with same model configuration and data. Sensitivity and specificity refer to deidentification sensitivity and deidentification specificity, respectively. All values are reported with 95% confidence intervals. CI denotes confidence interval; FNR false-negative rate and FPR, false-positive rate.



Supplementary Figure 41: Pairwise bootstrap comparison of deidentification sensitivity per modality between the Multimodal Anonymizer and baseline systems. Each cell displays the mean difference in deidentification sensitivity (Δ Sensitivity) with 95% confidence intervals. Positive values (green) indicate superior performance of the Multimodal Anonymizer (MA) over the respective baseline. Results for table, text, image, PDF, metadata, and filenames were derived from automated evaluation, while scores for GRASCCO, handwriting, German letters, and faces stem from manual evaluation.



Supplementary Figure 42: Pairwise bootstrap comparison of deidentification specificity/precision per modality between the Multimodal Anonymizer and baseline systems. Each cell displays the mean difference in deidentification specificity or precision (Δ Specificity/Precision) with 95% confidence intervals. Positive values (green) indicate superior performance of the Multimodal Anonymizer (MA) over the respective baseline. For automatically evaluated modalities (table, text, images, PDF, metadata, filenames), specificity is reported, as true negatives are available. For manually evaluated modalities (GRASCCO, handwriting, German letters, faces), precision is reported instead, as the absence of true negatives precludes specificity calculation.